BASE24-atm

NETHERLANDS BGC INTERCHANGE INTERFACE MANUAL

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PREFACE

The BASE24-atm Netherlands BGC (NBGC) Interchange Interface is a BASE24 product that provides an interface between a BASE24 acquirer or issuer system and Interpay. The term NBGC Interchange Interface is used to describe the BASE24 process that interfaces with Interpay.

Intended Audience

This document is primarily intended for staff responsible for the configuration, operation or maintenance of the NBGC Interchange Interface.

Users are expected to be familiar with the BASE24 configuration, operation and for certain functions, Tandem Systems Management. Users should consult the relevant Tandem or BASE24 documentation where appropriate.

This document can be used as an aid to training but readers are advised to attend the relevant training course(s).

Location of Information

A list of section headings and related page references are given in the Contents page. The headings are function oriented.

Section 1—Introduction provides an introduction to the architecture of the BASE24 system and the IAN.

Section 2—Processing Overview provides an overview of the functionality of the BASE24 NBGC Interchange Interface.

Section 3—Configuration and Files Maintenance describes the configuration requirements specific to the BASE24 NBGC Interchange Interface.

Section 4—Startup and Control documents the network control commands supported by the BASE24 NBGC Interchange Interface and describes the network management message processing.

Section 5—Transaction Message Handling describes the different transaction message flows between the IAN and the BASE24 NBGC Interchange Interface and provides further information on transaction processing.

Section 6—File Action Message Handling describes the file action message flows between the IAN and the BASE24 NBGC Interchange Interface.

Section 7—Cutover and Reporting documents the procedures for settlement with Interpay and describes the configuration of standard BASE24 interchange reporting.

Section 8—EMV Support describes the support within the BASE24 NBGC Interchange Interface for EMV transaction processing.

Appendix A—Response Code Translation presents tables describing the translation of response codes between BASE24-atm and the IAN.

Appendix B—File Formats defines the data structures used by the BASE24 NBGC Interchange Interface.

Appendix C—Log Messages describes the messages logged by the BASE24 NBGC Interchange Interface and provides guidance on their causes and the appropriate action to take.

Appendix D – Glossary explains the definitions of the terms used in this manual.

Related Documentation

BASE24 Documentation

- NET24-XPNET Network Control Operations Guide
- NET24-XPNET Network Control Command Reference Manual
- BASE24 CRT Access Manual
- BASE24 Base Files Maintenance Manual
- BASE24 Logical Network Configuration File Manual
- BASE24 Files and Record Structures Reference Manual
- BASE24 Tokens Manual
- BASE24 Transaction Security Manual
- BASE24 Transaction Security Services Processing Guide
- BASE24-atm EMV Support Manual
- BASE24-atm Files Maintenance Manual
- BASE24-atm Settlement and Reporting Manual
- BASE24-atm Transaction Processing Manual

Interpay Documentation

- B.I.M. Functional Specifications, Version 4.2a, dated 29 June 2004
- B.I.M. Functional Specifications, Addendum B, Version 1.0, dated 21 February 2006 – Improved TCP/IP Session Management
- B.I.M. Functional Specifications, Addendum D, Version 1.3, dated 31 March 2006 – Chip Based Debit (Maestro, PIN, NGG)

Other Documentation

- ISO Codes for the Representation of Currencies and Funds, ISO 4217-1981
- ISO Standard for Bank Card Originated Messages, ISO 8583 - 1987
- ISO Standard for Bank Card Originated Messages, ISO 8583 - 1993
- American National Standards Institute (ANSI) Standard X9.8 for PIN Management and Security
- American National Standards Institute (ANSI) Data Encryption Standard (DES) X9.9
- Europay/MasterCard/Visa (EMV): EMV 2000 - ICC Specifications for Payment Systems, version 4.0, dated December 2000

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1. INTRODUCTION

Interpay's Interbancair Autorisatie Netwerk (IAN) is the national switch for financial transaction processing in the Netherlands. The BASE24 Nederland BankGiroCentrale (NBGC) Interchange Interface connects a BASE24 network with Interpay's IAN.

The Bank Interchange Message (BIM) protocol describes the interface between the IAN and Interpay Member systems. The BIM protocol has recently been enhanced to version 4.2a. The NBGC Interchange Interface supports the application level interface defined in the latest version of the BIM protocol. Please refer to the ***BIM Functional Specifications*** for further details.

1.1 BASE24 SYSTEM ARCHITECTURE

The NBGC Interchange Interface is a BASE24 switch product that supports the application level interface between a BASE24 system and Interpay's IAN in the Netherlands.

1.1.1 System Overview

Figure 1-1 provides a simplified representation of the role of the NBGC Interchange Interface within the BASE24 environment

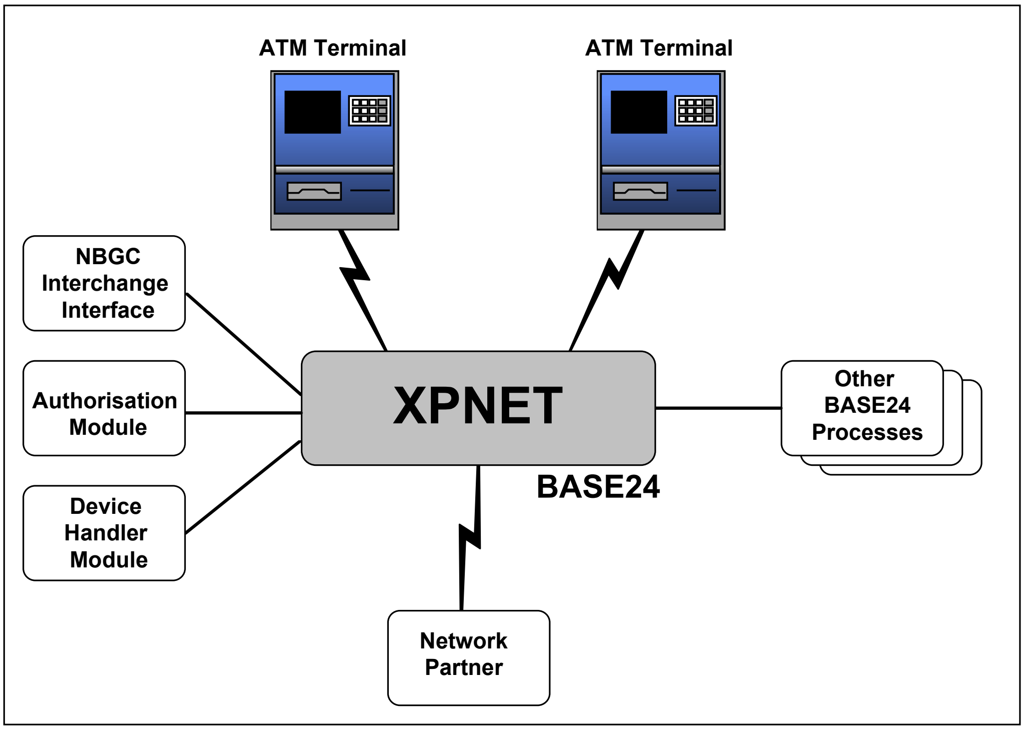

Figure 1-1 Role of the NBGC Interchange Interface within the BASE24 Environment

1.1.2 Issuer And Acquirer Support

In the financial industry, the terms *issuer* and *acquirer* define the parties on opposite ends of the electronic message exchanges required to authorise and process transactions.

- Issuers are card-issuing or account-owning institutions or their agents. These institutions or agents are responsible for authorising cardholder transactions and keeping track of cardholder accounts.
- Acquirers are the parties initiating transactions on behalf of the card acceptors.

The BASE24 NBGC Interchange Interface can function in the role of an issuer, an acquirer, or both. When the BASE24 NBGC Interchange Interface sends financial transactions to the IAN for authorisation, the BASE24 NBGC Interchange Interface is representing the transaction acquirer, even though the transaction may not have originated at a BASE24-controlled terminal. When financial transactions are being sent to the BASE24 NBGC Interchange Interface for authorisation from the IAN, the BASE24 NBGC Interchange Interface is representing the card issuer, even though BASE24 may only forward the transaction to an issuing system for authorisation.

1.2 IAN SYSTEM ARCHITECTURE

The IAN is a telecommunications network operated by Interpay. The network allows communications between Interpay Members for transaction authorisation. The network also acts as a “gateway” between Interpay Members and the international card schemes, i.e. MasterCard (via EPS-Net) and Visa International (via VisaNet).

1.2.1 Network Overview

Figure 1-2 provides an overview of the IAN configuration.

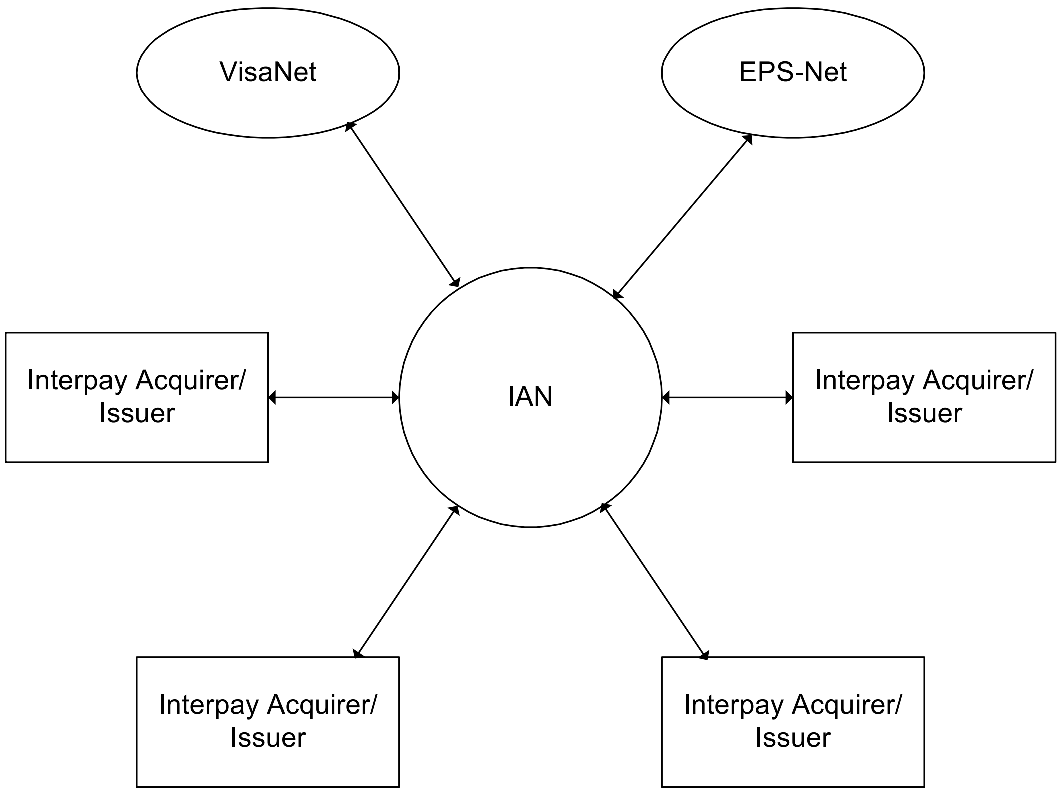

Figure 1-2 IAN Configuration

1.2.2 Switch Functions

The main function of the IAN Switch is message routing. Authorisation requests are sent from Acquirer Host to Issuer Host for authorisation and the result is sent back from Issuer Host to Acquirer Host. All processing is performed online and in real time.

In addition, the IAN provides 'stand-in' facilities to card issuers. An issuer may give permission to the Switch to authorise on behalf of the issuer should the Issuer Host be unavailable. This service is known as IAN stand-in.

1.2.3 BIM Specification

The Bank Interchange Message (BIM) protocol describes both the data communications and the application level interface:

- The data communications interface handles the transport of data. The interface supports the SNA LU0, X.25 and TCP/IP protocols. The BIM Specification defines the requirements for the physical, data link, and network layers of the data communication interface.

- The application level interface specifies the dialogues and message formats. The messages are based on the ISO Standard for Bank Card Originated Messages, ISO 8583-1987 and ISO 8583-1993. The BIM Specification defines the requirements for message processing between Interpay Members and the IAN.

For a full description of the data communication interface between a member's host system and the IAN, please refer to Section 3 of the ***BIM Functional Specifications.***

For a full description of the application level interface between a member's host system and the IAN, please refer to Section 4 of the ***BIM Functional Specifications.***

1.3 IAN MESSAGE PROCESSING

The interaction between Interpay Members and the IAN is accomplished by means of message exchange.

The IAN uses different types of message to support the following application functions:

- Network Management Function;
- Authorisation Function;
- File Action Function.

This message format is based on standards (8583–1987 and 8583-1993) developed by the International Organisation for Standardisation (ISO). It is a variable-length, variable-content message that contains different information, based on the type of message being sent.

1.3.1 Message Types

The IAN supports the following message types:

- 0100 Authorisation request
- 0110 Authorisation request response
- 0200 Financial Transaction request
- 0210 Financial Transaction request response
- 0220 Financial Transaction advice
- 0230 Financial Transaction advice acknowledgment
- 0420 Acquirer Reversal advice
- 0430 Acquirer Reversal advice acknowledgment
- 0800 Network management request
- 0810 Network management request response
- 0820 Network management advice
- 0830 Network management advice acknowledgment
- 1304 File Action request
- 1314 File Action request response
- 1324 File Action advice
- 1334 File Action advice acknowledgment
- 1624 Administrative advice
- 1634 Administrative advice acknowledgment

The 13xx message types are used to support the File Action function. The 08xx message types are used to support the Network Management function. The other message types are used to support the Authorisation function.

1.3.2 Network Management Function

The Network Management function provides transport link maintenance and support for session establishment, maintenance and disconnection.

Two types of Network Management message are defined:

- The **Network Management Request** is used by a member and the IAN to inquire about the presence of the other communication partner (echo testing).
- The **Network Management Advice** is used by the IAN to logoff from a member, and by a member to support the following functions:
  - Logon. The purpose of the logon dialogue is to establish a session between the IAN and the member, and to exchange new cryptographic keys.
  - Logoff. The purpose of the logoff dialogue is to terminate a session between the IAN and the member.
  - Issuer down. The purpose of the issuer down dialogue is to (temporarily) sign-off a member as issuer.
  - Issuer up. The purpose of the issuer up dialogue is to sign-on a member as issuer again.

For more details on the Network Management function, please refer to Section 4.4 of the ***BIM Functional Specifications.***

1.3.3 Authorisation Function

The Authorisation function provides 'online-to-issuer' authorisation for card-originated transactions. The interaction between acquirer and issuer constitutes an application. Authorisation function dialogues differ between authorisation applications. Each application supports its own application dialogue, comprising one or more transaction types, and two or more message types.

For more details on the Authorisation function, please refer to Section 4.5 of the ***BIM Functional Specifications.***

1.3.3.1 Application Types

The IAN supports the following application types:

- ATM;
- EFT/POS;
- Chipknip (electronic purse);
- Electronic commerce;
- MOTO;
- Top-Up.

1.3.3.2 Transaction Types

The IAN supports the following transaction types:

- Cash Withdrawal
- Deposit
- Balance Enquiry
- Purchase of Goods and Services
- Pre-authorisation
- Cash advance
- Refund
- Verification
- Electronic Purse Load
- Pre-pay Mobile Phone Top-Up

1.3.3.3 Application Dialogues

Each application supports a subset of the message types and transaction types listed above, as shown in the table below.

Application Type	Supported Transaction Types	Supported Message Types
ATM	Cash Withdrawal Deposit Balance Enquiry	0200/0210/0420/0430 0100/0110/0420/0430 0100/0110
EFT/POS	Purchase Pre-Authorisation Cash Advance Refund Deposit Verification	0200/0210/0420/0430 0100/0110/0220/0230/0420/0430 0200/0210/0420/0430 0200/0210/0420/0430 0200/0210/0420/0430 0100/0110
Chipknip	Purse Load	0200/0210/0420/0430
e-Commerce	Purchase	0200/0210/0420/0430
MOTO	Purchase	0200/0210/0420/0430
Top-Up	Top-Up	0200/0210/0420/0430

The BIM protocol also supports administrative advices (i.e. 1624 messages) to forward chip card generated notices (called Informational Advices) to the card issuer. Administrative advice acknowledgements (i.e. 1634 messages) are sent in response to administrative advice messages.

The BIM protocol also supports advices of transactions (i.e. 0220 messages) authorised in stand-in by MasterCard Europe.

1.3.3.4 EMV Support

The IAN supports the processing of transactions performed on EMV-compliant chip cards. The BIM protocol accommodates the transmission of EMV-related transaction information in a specific data element within the ISO 8583 message format.

1.3.4 File Action Function

The File Action function provides authorising entities (either card issuer or Stand-in/On-behalf) the capability of changing the status of a card. The entity may wish either to block or to unblock a card. The purpose of blocking a card is to restrict the card usage. The purpose of unblocking a card is to remove those restrictions.

Two types of File Action message are defined:

- The **File Action Request** is used to warn/advise the card issuer to block a card. Such requests can only be initiated by Stand-in/On-behalf to inform the card issuer of excessive PIN tries and by the BankPassenMeldCentrale in case a card has been reported lost or stolen by the cardholder.
- The **File Action Advice** is used to direct the Stand-in/On-behalf system to block or unblock a card.

For more details on the File Action function, please refer to Section 4.6 of the ***BIM Functional Specifications.***

1.3.5 Network Security

The security concept within the IAN is based on zone security. Transition from one zone to another is at the Switch level. The following descriptions apply to a single zone, i.e. between the Switch and an acquirer and/or issuer system.

To ensure the receiver of a message is able to determine its authenticity and that the information sent across the network cannot be fraudulently manipulated without detection, the following principles apply:

- A MAC is added to the message to protect the authenticity of the message and the sender. The message originator passes the MAC to the message destination in ISO data element S-128. The MAC is calculated over all data elements within the message.
- PIN transport is protected by encrypting the PIN from the moment the PIN is entered at the point of service to the moment of PIN verification. This ensures a PIN never appears in plain text outside the secure environment of the Host Security Module (HSM).
- Encryption of data other than the PIN is done at data link level by means of hardware encryption/decryption devices.

The security measures above are based on the use of the Data Encryption Standard (DES) using secret keys. Since April 1, 2003 only Triple DES has been used in the IAN; Single DES is no longer supported.

The BIM protocol also provides for online key exchange.

For a full description of the IAN security processing, please refer to Section 5 of the ***BIM Functional Specifications.***

1.3.5.1 MAC Session Keys

The same MAC Session Key is used for acquirer and issuer traffic. When sending a message to the IAN, a MAC is generated using the MAC session key, and it is this value that is carried in S-128. When receiving a message from the IAN, the MAC is retrieved from S-128, and verified using the same MAC Session Key.

The MAC Session Key is dynamically exchanged between the institution and the IAN by using network management messages, and is renewed when a new session is started.

1.3.5.2 PIN Session Keys

The same PIN Session Key is used for acquirer and issuer traffic. When sending a request message to the IAN, the encrypted PIN Block is retrieved from the internal message. The PIN Block is translated to encryption under the PIN Session Key, and it is this encrypted value that is carried in P-52. When receiving a request message from the IAN, the PIN Block is retrieved from P-52, encrypted under the same PIN Session Key.

The PIN Session Key is dynamically exchanged between the institution and the IAN by using network management messages, and is renewed when a new session is started.

1.4 IMPLEMENTATION CONSIDERATIONS

This section documents environmental issues to be considered when implementing the NBGC Interchange Interface.

1.4.1 Software Dependencies

The NBGC Interchange Interface is compatible with XPNET version 3.0.

The product is a Switch 6.0 interface, and therefore operates in a Release 3.4/4.0, a Release 5.0, a Release 5.1, a Release 5.3, or a Release 6.0 BASE24 system.

To provide full support for the encryption of PIN blocks using Triple DES, the Transaction Security Services (TSS) must be installed and configured.

To support acquirer surcharging, the BASE24 Surcharge product must be licensed and installed.

In order to support the Dutch-specific tokens (identifiers X0, X1, X2), the NBGC Interchange Interface must be installed and configured in a BASE24 systems containing the country-specific product code for the Netherlands (subvolume BA60NLCS).

1.4.2 Disk File Requirements

Each NBGC Interchange Interface process writes records to an Interchange Log File (ILF) and a Store-and-Forward File (SAF). Sufficient disk space must be available for NBGC Interchange Interface processes to write the required information to each of these files.

1.4.3 Security Requirements

The minimum device and firmware requirements for HSMs in order to support Triple DES in the interface to the IAN are as follows:

- Thales e-security (formerly Racal) HSMs must be RG7000 with Revision 1.05/5.05 firmware or greater.
- The Thales HSMs must also support the Dutch-specific security commands. Atalla HSMs cannot be used, because Atalla does not support the Dutch-specific security commands.

1.4.4 EMV Support

To support EMV processing, the BASE24-atm EMV product must be licensed and installed.

2. PROCESSING OVERVIEW

The NBGC Interchange Interface process is responsible for the online message transfer between BASE24 and the IAN.

This process performs the following functions:

- Manages communications between BASE24 and the IAN.

  The NBGC Interchange Interface sends and receives network management messages to determine the status of the link between BASE24 and the IAN.

- Facilitates transaction message exchange between BASE24 and the IAN.

  When BASE24 receives a transaction from the IAN, the NBGC Interchange Interface translates the message into the appropriate internal message format and sends the message to a BASE24 Authorisation process. When BASE24 receives a transaction for the IAN to authorise, the NBGC Interchange Interface translates the message from the internal message format to the external message format and forwards the message to the IAN. The NBGC Interchange Interface supports communication with the IAN using the message formats defined in the ***BIM Functional Specifications.*** These formats are based on the ISO 8583 (1987) message standard.

- Provides store-and-forward capabilities.

  When the link between BASE24 and the IAN is down, the NBGC Interchange Interface stores certain messages intended for the IAN in its Store-and-Forward File (SAF). When the link is restored, these messages are retrieved from the file and sent to the IAN.

- Supports card status updates between BASE24 and the IAN via online message exchange.

  BASE24 can receive a File Action Advice from the IAN, and can send a File Action Advice to the IAN. The NBGC Interchange Interface supports communication with the IAN using the message formats defined in the ***BIM Functional Specifications.*** These formats are based on the ISO 8583 (1993) message standard.

- Facilitates settlement between BASE24 and the IAN.

  The NBGC Interchange Interface produces a set of interchange reports to assist in reconciling transactions between BASE24 and the IAN. These reports allow the BASE24 participants to reconcile the settlement positions of the IAN and BASE24.

2.1 FILE USAGE

The NBGC Interchange Interface process uses a number of BASE24 files. These files, which can be divided into control files and transaction files, are identified below.

2.1.1 Control Files

Control files contain configuration information and parameters that determine how the NBGC Interchange Interface process operates. The control files used by the NBGC Interchange Interface include:

- Network Environment File (NEF)
- Logical Network Configuration File (LCONF)
- Interchange Configuration File (ICF)
- Key File (KEYF or KEY6)
- Log Message Configuration File (LMCF)
- Token File (TKN)
- External Message File (EMF)

For information on how these files are used to configure the NBGC Interchange Interface process, refer to section 3.

For information on the switch-specific ICF data definition, refer to Appendix B.

2.1.2 Transaction Files

Transaction files are used to store information about individual transactions processed through the NBGC Interchange Interface. The transaction files used by the NBGC Interchange Interface include:

- Interchange Log File (ILF)
- Store-and-Forward File (SAF)

The ILF and SAF DDLs can be accessed using DDLDOC. For information on DDLDOC, refer to the ***BASE24 Files and Record Structures Reference Manual***.

2.2 NETWORK MANAGEMENT

The BASE24 NBGC Interchange Interface connects a BASE24 system with the IAN. The communications between the networks are managed by ACI's XPNET software and application-level network management messages.

For further information on network management processing in the NBGC Interchange Interface, refer to section 4.

2.2.1 XPNET

Stations provide the means by which communications takes place between BASE24 and the IAN. All messages sent to the IAN Switch are actually sent to the stations defined for the IAN. Likewise, all messages received from the IAN are actually received from stations.

All protocol-level functions between BASE24 and the stations are handled by XPNET using the communications control software, provided with the HP NonStop (formally Tandem) hardware. The NBGC Interchange Interface process sends and receives messages through XPNET. It is the responsibility of XPNET to maintain the communications links with the stations according to the protocols under which the stations are defined.

Configuration of an XPNET system includes defining attributes for each XPNET node and all of its associated processes, links, lines, stations, and devices. All objects, including the XPNET node itself, can be added, changed, and deleted online without stopping the XPNET system. Configuration information relating to each XPNET node is held in the node's Network Environment File (NEF). There is one NEF for each XPNET node in the system. Each XPNET process manages the NEF in that XPNET node.

No two XPNET systems are configured exactly alike. In addition to being composed of different applications, processes, devices, and stations, there are a number of settings that depend entirely on how you want your system to work.

For complete information on the syntax of the commands, attributes, and directives used to configure an XPNET system, see the ***NET24-XPNET Network Control Command Reference Manual.***

2.2.2 Message Types

The NBGC Interchange Interface module supports network management message types, as shown in the following table.

Type	Description
0800	Network management request
0810	Network management request response
0820	Network management advice
0830	Network management advice acknowledgement

The structure and content of the external messages used by the NBGC Interchange Interface process to communicate with the IAN are specified in the ***BIM Functional Specifications***.

Network Management messages are used to support the following functions:

- Echo Test
- Logon
- Logoff

Network Management messages are also used to support dynamic key exchange. Session Keys and Transport Keys are exchanged in logon messages.

The NBGC Interchange Interface module also supports Probe message types, to provide improved TCP/IP session management, as described in the ***B.I.M. Functional Specifications, Addendum B, Improved TCP/IP Session Management.***

2.3 TRANSACTION PROCESSING

The BASE24 NBGC Interchange Interface connects a BASE24 system with the IAN. It allows an Interpay cardholder to initiate a transaction at a terminal belonging to a network not operated by the card issuer.

For further information on transaction processing in the NBGC Interchange Interface, refer to section 5.

2.3.1 Message Formats

This section defines supported message types and the translation between the external and internal message formats.

2.3.1.1 Supported Message Types

The NBGC Interchange Interface module supports both acquirer and issuer processing for the ATM application, and issuer processing only for other authorisation applications. A single process can process messages associated with any of the authorisation applications supported by the NBGC Interchange Interface.

The following table indicates the transaction message types supported by the NBGC Interchange Interface.

Type	Description
0100	Authorisation request
0110	Authorisation request response
0200	Financial transaction request
0210	Financial transaction request response
0220	Financial transaction advice
0230	Financial transaction advice acknowledgement
0420	Acquirer reversal advice
0430	Acquirer reversal advice acknowledgement
1624	Administrative advice (receiver only)
1634	Administrative advice acknowledgement (sender only)

The structure and content of the external messages used by the NBGC Interchange Interface process to communicate with the IAN are specified in the ***BIM Functional Specifications***.

2.3.1.2 Message Translation

Within BASE24, transaction processing uses an internal message format. The internal message format is used whenever information is routed between the NBGC Interchange Interface process and the rest of BASE24. The internal message is commonly referred to as the BASE24-atm Standard Internal Message (STM). For more information on the STM, see the ***BASE24-atm Transaction Processing Manual***.

The NBGC Interchange Interface process translates request messages it receives from the IAN into the internal message format, and sends the internal request to the BASE24-atm Authorisation process. When the message has been processed, the BASE24-atm Authorisation process sends an internal response to the NBGC Interchange Interface process. Using information from the internal message, the NBGC Interchange Interface process formats an external response message and sends it to the IAN.

The NBGC Interchange Interface process translates the internal request message it receives from BASE24-atm into the external message format, and sends the external request to the IAN for authorisation. When the message has been processed, the IAN sends an external response to the NBGC Interchange Interface process. Using information from the external message, the NBGC Interchange Interface process formats an internal response message and sends it to the BASE24-atm Authorisation process.

2.3.2 Supported Transaction Types

- The NBGC Interchange Interface forwards the following BASE24-atm transactions to the IAN for authorisation:
- Balance Enquiry
- Cash Withdrawal

The NBGC Interchange Interface may receive the following transaction types from the IAN for authorisation:

- Balance Enquiry
- Cash Withdrawal
- Purchase
- Pre-authorisation
- Load transaction for electronic purse

When a purchase, pre-authorisation, or electronic purse load transaction is received from the IAN, the processing code is mapped to a BASE24-atm cash withdrawal transaction.

2.3.3 Transaction Logging

The NBGC Interchange Interface logs the following types of messages to the Interchange Log File (ILF):

- Requests containing error conditions
- Responses
- Reversals
- Force posts

Whenever a reversal occurs, the NBGC Interchange Interface attempts to update the ILF record already logged for the transaction.

2.3.4 Store-and-Forward Function

A store-and-forward function is incorporated into the NBGC Interchange Interface process to collect transaction messages during periods when the IAN is unavailable. When the IAN is unavailable, messages required as notification to the IAN are stored in the Store-and-Forward File (SAF). These messages are then transmitted later when communications are restored.

2.3.5 Transaction Security

The transaction acquirer passes the PIN data to the card issuer in ISO data element P-52. The BASE24 NBGC Interchange Interface supports the transmission of PIN Blocks encrypted using Triple DES only (Single DES encryption is no longer permitted).

Triple DES is supported fully only in BASE24 Release 6.0 (the full solution). To facilitate a phased migration, BASE24 Switch 6.0 interchange interfaces can be used in a 5.x environment (and also in Release 6.0) to provide limited Triple DES support (the interim solution).

2.3.5.1 Triple DES “Full” Solution

With the advent of Triple DES, ACI has introduced a new architecture for managing transaction security. In BASE24 Release 6.0, all cryptographic functions, such as PIN translation, can be managed through the new *BASE24 Transaction Security Services*. A full Triple DES environment is based on BASE24 6.0 using the Transaction Security Services.

In the traditional BASE24 environment, the application processes (device handlers, authorisation modules and interface processes) manage their own cryptographic services. This includes storing and retrieving keys, and making calls to a Hardware Security Module (HSM). In an environment using Transaction Security Services (TSS), the applications call the TSS process rather than the HSM. TSS manages the storing and retrieving of keys and making calls to the HSM.

When using BASE24 Transaction Security Services, the application processes use “key locators” in place of actual keys. Key locators serve as tags, or indexes, representing a key. The actual keys are stored in new key files managed exclusively by Transaction Security Services. The TSS uses the key locators to retrieve the actual keys, before formatting and sending commands to the HSM.

The full solution impacts the NBGC Interchange Interface because of the new configuration requirements. The Key File (KEYF) stores key locators to the TSS database instead of the actual PIN block encryption keys themselves.

Although designed to support Triple DES, the TSS also supports Single DES.

Further details of the TSS are given in the ***BASE24 Transaction Security Services Processing Guide***.

2.3.5.2 Triple DES "Interim" Solution

Transaction Security Services is not supported prior to Release 6.0, however, the same version of BASE24 interchange interfaces can run in all releases of BASE24. To enable Release 6.0 interchange interfaces to run in all releases of BASE24, modifications have been made to interchanges interfaces to support Triple DES when the interface is running in a BASE24 Release 5.x network. The interchange interfaces must support translating a PIN block in transactions originating at the interchange from Triple DES encryption to Single DES encryption. The interchange interface must support translating a PIN block in transactions originating in BASE24 or a host from Single DES encryption to Triple DES encryption. To accomplish this, the interchange interfaces call utilities added to support double length PIN encryption keys in PIN block translation calls to an HSM. An LCONF parameter indicates when the special PIN block translation occurs.

This interim solution impacts the NBGC Interchange Interface in the following ways:

- Support for the new KEY6 file, replacing the Key File (KEYF). This new file stores double length PIN block encryption keys instead of the existing single length PIN block encryption keys.

- Support for translating PIN blocks encrypted using Single DES (received from the BASE24-atm Authorisation process) to encryption using Triple DES (for transmission to the IAN).

- Support for translating PIN blocks encrypted using Triple DES (received from IAN) to encryption using Single DES (for transmission to the BASE24-atm Authorisation process).

An additional call to the HSM (to convert a PIN Block from Triple DES encryption to Single DES encryption) will be required to support PIN Blocks received from the IAN if the Triple DES "interim" solution is implemented.

2.3.6 Transaction Routing

This section outlines how BASE24 routes transaction messages to and from the NBGC Interchange Interface.

2.3.6.1 Outbound Request Routing Method

Transactions acquired by BASE24 and destined for the IAN are routed to the NBGC Interchange Interface via the XPNET gateway process.

The Authorisation process identifies the card prefix for the request and, when unable to locate a matching entry in the Card Prefix File (CPF), forwards the request to the gateway for onward routing.

The gateway determines the required destination for the transaction request according to routing criteria held in the routing files, SPREFIX and PREMAP. The PREGEN configuration process is used to create SPREFIX and PREMAP. PREGEN uses the BASE24 Prefix Files (Interchange Prefix File and CPF), the SPANCNTL file and a PREGEN command file as input to the process.

A variety of routing methods are used by the gateway. For more information, refer to the ***BASE24 Configuration Manual***.

The routing logic for each message routed by the gateway is described in terms of:

- Message source (logical network and process name)
- Target destination (logical network and process name)
- Routing method.

Figure 2-1 is an example of the routing criteria that is appropriate for an ATM system with one logical network and a single NBGC Interchange Interface process.

It indicates that requests originating from any process in the logical network (PRO1) are routed to the NBGC Interchange Interface (P1A^NBGC) using the Logical Network Prefix Look Up method, where the valid prefixes are contained in the IPF named IPF1. If the gateway finds a match in the SPREFIX file entry built from IPF1, then the transaction will be routed to the specified destination, otherwise the transaction will be returned to its source as undeliverable.

```
SOURCE     PAIR        ORDERED     ROUTING       Prefix File Name
LN         Process     Method      Dest

PRO1       *           LGNT        P1A^NBGC      \SYS1.$DATA.PRO1DATA.IPF1
```

Figure 2-1 Example Routing Criteria

Prefixes are added to a named IPF using the AFT PREFIX BUILD screens. (File destination 'PRE'.)

2.3.6.2 Inbound Requests

Transactions received from the IAN that are inbound to BASE24, are routed by the NBGC Interchange Interface process to the ATM Authorisation process named in the ICF record.

2.4 FILE ACTION PROCESSING

The BASE24 NBGC Interchange Interface connects a BASE24 system with the IAN. It allows authorising entities to change the status of a card, using application-level file action messages.

For further information on file action processing in the NBGC Interchange Interface process, refer to section 6.

2.4.1 Message Types

The NBGC Interchange Interface module supports file action message types, as shown in the following table.

Type	Description
1324	File action advice
1334	File action advice acknowledgement

The structure and content of the external messages used by the NBGC Interchange Interface process to communicate with the IAN are specified in the ***BIM Functional Specifications***.

File Action messages are used to indicate which action should be taken on the card file with respect to a specific card. The NBGC Interchange Interface module supports the following file action functions:

- Add record;
- Delete record.

These functions allow the card status to be changed from unblocked to blocked, or from blocked to unblocked.

A customer can configure whether the NBGC Interchange Interface logs file action messages to the Interchange Log File (ILF).

2.4.2 Message Translation

File Action messages are not actioned by BASE24, and are not initiated by BASE24:

- When a File Action advice is received from the IAN, the File Action message is forwarded to a Host system, via a BASE24 Host Interface process.
- A File Action advice is sent to the IAN, only when a File Action message has been received from a Host system, via a BASE24 Host Interface process.

File Action messages are passed between the NBGC Interchange Interface and the Host Interface in a format used within BASE24 for file action messages. This format is commonly referred to as the BASE24 Standard External Message (SEM). For more information on the SEM, see the ***BASE24 External Messages Manual***.

The NBGC Interchange Interface process translates advice messages it receives from the IAN into the SEM format, and sends the SEM to the Host Interface process. Without waiting for a response, the NBGC Interchange Interface process formats an external acknowledgement message and sends it to the IAN.

The NBGC Interchange Interface process translates the SEM it receives from the Host Interface into the external message format, then sends the advice message to the IAN. The IAN sends an external acknowledgement to the NBGC Interchange Interface process.

2.5 INTERCHANGE REPORTING

The NBGC Interchange Interface produces two reports: SETL01 and STAT09. The purpose of these reports is to facilitate settlement between BASE24 and the IAN.

For information on these reports, refer to section 7.

2.6 CHIP-BASED DEBIT/CREDIT SUPPORT

The IAN supports chip-based debit and credit functionality, based on the EMV Standards: ***EMV 2000 - ICC Specifications for Payment Systems***.

For authorisation systems such as BASE24, this involves supporting additional data elements in the BIM protocol and supporting additional values in existing data fields.

EMV support is provided by a separately licensable module, which is built on top of the core NBGC Interchange Interface modules.

For information on the EMV extension module, refer to section 8.

Full details of EMV support in BASE24 are provided in the ***BASE24-atm EMV Support Manual***.

2.7 UNSUPPORTED FUNCTIONALITY

The following restrictions apply to message processing in the NBGC Interchange Interface:

- On a BASE24 acquirer system, the NBGC Interchange Interface supports transaction messages for the ATM application only.
- On a BASE24 issuer system, the NBGC Interchange Interface supports transaction messages for the ATM, EFT/POS and Chipknip applications only.
- On a BASE24 issuer system, the NBGC Interchange Interface supports processing codes of 000000 (purchase), 010000 (cash withdrawal), 070000 (pre-authorisation), 140000 (purse load), 310000 (balance inquiry) only.
- On a BASE24 system (acquirer and issuer), the NBGC Interchange Interface does not support the sending of Administrative Advice (1624) messages or the receiving of Administrative Advice Acknowledgement (1634) messages.
- On a BASE24 system (acquirer and issuer), the NBGC Interchange Interface does not support File Action Request/Response (1304/1314) messages.
- On a BASE24 system (acquirer and issuer), the NBGC Interchange Interface does not support the “Change Record” and “Replace Record” functions in File Action Advice/Acknowledgement (1324/1334) messages.
- On a BASE24 system (acquirer and issuer), the NBGC Interchange Interface does not support the “Issuer Up/Issuer Down” functions in Network Management Advice/Acknowledgement (0820/0830) messages.

Support for the following functions may be added in a future release of the NBGC Interchange Interface:

- Add support for field P-48.71 in request messages received from Interpay, as described in ***B.I.M. Functional Specifications, Addendum D, Chip Based Debit (Maestro, PIN, NGG)***;
- Add support for version 4.3a of the BIM Functional Specifications, specifically “Wellink Optimaal Scenario”;
- Resolve issues regarding the format of data in the ILF, e.g. redefinitions of user fields, dynamic allocation of ILF record space;
- Rationalise the use of the NBGC Reversal Token;
- Rationalise the use of the NBGC Transaction Token.

3. CONFIGURATION AND FILES MAINTENANCE

This section describes the configuration files that must be set up or verified in order that the NBGC Interchange Interface process can be configured into the BASE24 network.

This section also provides details of the BASE24 files maintenance screens and field descriptions that have been enhanced to support the BASE24 NBGC Interchange Interface module. Function keys that operate differently from the standard function keys are also documented.

For full details of the screen layouts, standard BASE24 function keys and online help screens, refer to the ***BASE24 Base Files Maintenance Manual***.

3.1 XPNET PROCESS

The XPNET process supports the following data communications protocols for connection to the IAN:

- X.25
- TCP/IP
- SNA LU0

A single BASE24 NBGC Interchange Interface process can support a maximum of four stations with the IAN.

This section provides a brief explanation of the way in which BASE24 lines, stations and devices are defined for the NBGC Interchange Interface process.

See the ***NET24-XPNET Network Control Operations Guide*** for descriptions of the XPNET network control functions. This manual provides descriptions of the various functions used in configuring, monitoring, and controlling an XPNET system.

3.1.1 Terminology of Lines and Stations

All BASE24 communication to external devices takes place using logical entities known as stations and lines. For asynchronous and byte synchronous protocols, a line is synonymous with a physical communications line and there can be one or more stations per line, depending upon the protocol. For X.25, however, a line is equivalent to an X.25 virtual circuit and only one station can be defined per line.

3.1.2 Network Configuration

All processes, lines, devices, and stations must be defined in the Network Environment File (NEF) before they can be recognised. This means that any processes, lines, devices or stations specific to the NBGC Interchange Interface must also be defined in the NEF, as described below.

- Process. There must be one process declaration for each NBGC Interchange Interface process.
- Line. There must be one line declaration for each communication line used by the NBGC Interchange Interface process.
- Device. There must be one device declaration for each generic station type used by the NBGC Interchange Interface process.
- Station. There must be one station declaration for each station in the network used by the NBGC Interchange Interface process.

See the ***NET24-XPNET Network Control Command Reference Manual*** for descriptions of the XPNET network control commands. This manual provides comprehensive information on configuring processes, lines, devices and stations in the NEF.

3.2 PREGEN CONFIGURATION

PREGEN must be run to create the initial version of SPREFIX and PREMAP. The PREGEN command file (PRECMD) is used as the command file for this operation.

All PREGEN command files reside in the $SYSTEM.<Span System Name>CNTL subvolume.

The obey file, GOPRE, runs PREGEN for an initial or next build function. This is normally run as part of the XPNET configuration sequence.

An example of the PRECMD file for the Initial Build is given below.

```
!********************************************************************
!
!                  PREGEN COMMAND FILE (PRECMD)
!
!SOURCE   PAIR    ORDERED   ROUTING         Prefix File Name
!LN               Process   Method          Dest
!
!********************************************************************
!
PRO1       *      LGNT      P1A^NBGC        \SYS1.$DATA.PRO1DATA.IPF1
!
*********************************************************************
BUILDNEXT;
BUILDEND;
SPANCNTLUSE        $SYSTEM.PRODCNTL;
SPANCNTLEND;
PREMAPCREATE;
PREMAPEND;
PRADD              PRO1,  *,  LGNT,  P1A^NBGC,  $DATA.PRO1DATA.IPF1,  IP;
PREND;
BUILDSPREFIX;
SPREFIXEND;
```

PREGEN can also be run to add prefixes to the current version.

For banks to accept additional card prefixes to be forwarded to NBGC, a new version of the system is not required. The new prefixes are added to the IPF, and the PRECMDCR command file is used to update the current version. When this mode is used to replace the SPREFIX file for a current configuration, only PREGEN and DISTRIB are run. The new SPREFIX file can be used by the gateway following a warmboot via the NCS SPANCNTL screen.

The obey file, GOPRECR, runs PREGEN to update a current configuration and the obey file, GODISTCR, distributes an update to the current configuration.

An example of the PRECMDCR file is given below.

```
!********************************************************************
!
!                  PREGEN COMMAND FILE (PRECMDCR)
!
*********************************************************************
BUILDCURRENT;
BUILDEND;
SPANCNTLUSE        $SYSTEM.PRODCNTL;
SPANCNTLEND;
PREMAPUSE;
PREMAPEND;
BUILDSPREFIX;
SPREFIXEND;
```

3.3 LOGICAL NETWORK CONFIGURATION FILE (LCONF)

The Logical Network Configuration File (LCONF) is the central location of file information and processing information for BASE24. One LCONF exists for each logical network.

The LCONF contains assigns that link the physical names of objects in the logical network to names used in the BASE24 code and parameters (params) that define processing values at the logical network level.

For more details, refer to the ***BASE24 Logical Network Configuration File Manual***.

The LCONF assigns and params listed in this subsection are required by the NBGC Interchange Interface to initialise and process correctly.

3.3.1 LCONF Assigns

Unless otherwise stated, the assigns specified below are optional.

EMF

The fully qualified file name of the External Message File (ICF); for example, \B24.$DATA.PRO1DATA.EMF.

ICF

The fully qualified file name of the Interchange Configuration File (ICF); for example, \B24.$DATA.PRO1DATA.ICF. Required.

ILF

The fully qualified file name of the Interchange Log File (ILF); for example, \B24.$DATA.PRO1DATA.ILYYMMDD. Required.

Process Name	Required (for example, P1A^NBGC) because the ILF assign must be associated with the symbolic name of the NBGC Interchange Interface process.
Template File	Required (for example, \B24.$DATA.PRO1TPLT.ILF) because the ILF is created periodically by BASE24. The template file is used by the BASE24 NBGC Interchange Interface to determine the extent sizes and file characteristic for the ILFs, which are created daily.

KEY6

The fully qualified file name of the Key File (KEY6) required for the Triple DES "interim" solution; for example, \B24.$DATA.PRO1DATA.KEY6. Required if the Transaction Security Services (TSS) have not been implemented.

KEYF

The fully qualified file name of the Key File (KEYF) required for the Triple DES "full" solution; for example, \B24.$DATA.PRO1DATA.KEYF. Required if the Transaction Security Services (TSS) have been implemented.

LMCF

The fully qualified file name of the Log Message Configuration File (LMCF); for example, \B24.$DATA.PRODCNTL.LMCF. The LMCF assign is used by all of the application processes and, optionally, any XPNET that can alter log messages.

SAF

The fully qualified file name of a Store And Forward File (SAF); for example, \B24.$DATA.PRO1DATA.SAF1. Required.

Process Name	The SAF assign must be associated with the symbolic name of the NBGC Interchange Interface (for example, P1A^NBGC) because the dates of the SAFs may vary from process to process.
Template File	Required (for example, \B24.$DATA.PRO1TPLT.SAF) because the SAF is created periodically by BASE24.

TKN

The fully qualified file name of the Token File (TKN); for example, \B24.$DATA.PRO1DATA.TKN. Required.

For assigns required for Hardware Security Module (HSM) use, refer to the ***BASE24 Transaction Security Manual***.

3.3.2 LCONF Params

Unless otherwise stated, the parameters specified below are optional.

BASE24-REL

A code indicating whether the BASE24 logical network is currently running under release 3.4/4.0 or release 5.x or release 6.0. Valid values are '40', '50' and '60'.

Param Default 60.

DES-TRIPLE-SINGLE

This optional parameter indicates if Triple DES support in the interface is provided using the "interim" solution, i.e. with the KEY6 (Transaction Security Services not used). Valid values are 'Y' (Yes, KEY6 is used) and 'N' (No, KEYF is used).

Param Default N.

ILF-NOTIFY-INTERVAL

The number of writes made to the Interchange Log File (ILF) between calculating the percentage of file full. Each time a write is made to the ILF, a counter is increased by one. When the counter is equal to the value specified in this parameter, a calculation is performed to determine how full the ILF is. The counter is then reset to zero.

When the file reaches the percent full specified in the ILF-NOTIFY-PERCENTAGE parameter, a message is logged.

This parameter can be set to any numeric value from 10-32 766.

Param Default 50.

ILF-NOTIFY-PERCENTAGE

The percentage of file full at which a message is logged indicating the ILF is becoming full. The percentage of file full is calculated and this calculation is continued at the interval indicated in the ILF-NOTIFY-INTERVAL parameter, until more file space is made available.

Depending on the number and size of tokens logged to the file with the internal message, this calculation and the resulting log messages may not accurately reflect an ILF approaching capacity.

This parameter can contain any numeric value from 10 - 99.

Param Default 70.

ISOLATE-MISSED-PROBES

Maximum number of consecutive missed probes at which point the station should be isolated. The value of this parameter should be less than the value of the MAX-MISSED-PROBES parameter. The valid range is 1 – 98.

Param Default 1.

LN-CUTOVER

The time of day that the ATM logical network is cut over. All institutions in the logical network must be cut over by this time. This parameter can be set to any value, in 24-hour format. This is used by the NBGC Interchange Interface on initialisation to calculate the BASE24 posting date which will be used for transaction requests incoming from the NBGC Switch.

Param Default No default value. If this parameter is not present in the LCONF, the processes that access the LCONF for this parameter abend during initialisation.

MAX-MISSED-PROBES

Maximum number of consecutive missed probes at which point the station should be disconnected. The valid range is 1 – 99.

Param Default 4.

NBGC-AIR-FIELD-PREFIX

This parameter stores the prefix value that each NBGC Interchange Interface process uses to compose the unique Authorisation Identification Response (AIR). The AIR uniquely identifies every authorisation response sent from the NBGC Interchange Interface process to the issuer, in a given period.

This mandatory parameter must be set up for each NBGC process running on the BASE24 system. The value of this parameter is 1 character in length in the format 0-9 and A-Z.

Process Name Required (e.g. P1A^NBGC).

Param Default No default value. If this parameter is not present in the LCONF, the processes that access the LCONF for this parameter abend during initialisation.

NBGC-BAD-MAC-RESET-INTERVAL

This parameter indicates the interval (in minutes) after which the count of bad MACs maintained by the NBGC Interchange Interface is reset to zero.

Format of this param is MMMM.

Param Default 120.

NBGC-B24-BLOCKING-INST-ID

The Institution Identification Code of the BASE24 system obtained by the NBGC Interchange Interface. This is used to format the Transaction Originator Institution Identification Code (field S-94) in the 1324 File Action Advices (online blocking messages) to the IAN. The valid values contain between 1 and 11 numeric characters.

Param Default No default value. If this parameter is not present in the LCONF, the processes that access the LCONF for this parameter abend during initialisation.

NBGC-BGC-BLOCKING-INST-ID

The Institution Identification Code of the BGC Blocking system obtained by the NBGC Interchange Interface. This is used to format the Transaction Destination Institution Identification Code (field S-93) in the 1324 File Action Advices (online blocking messages) to the IAN. The valid values contain between 1 and 11 numeric characters.

Param Default No default value. If this parameter is not present in the LCONF, the processes that access the LCONF for this parameter abend during initialisation.

NBGC-BLOCKING-DPC

The Data Processing Centre (DPC) number of the Host system obtained by the NBGC Interchange Interface. The NBGC Interchange Interface formats 0300 File Update Requests to BASE24 specifying this DPC number. If the value of the NBGC-BLOCKING-HOST parameter is a Host Interface (HISO) process, the DPC number is used by the HISO process to access the Host Configuration File (HCF). The valid values are 0001 - 9999.

Param Default No default value. If this parameter is not present in the LCONF, the processes that access the LCONF for this parameter abend during initialisation.

NBGC-BLOCKING-HOST

The BASE24 logical name of a process obtained by the NBGC Interchange Interface. The NBGC Interchange Interface forwards 0300 File Update Requests to the specified process. Normally, this process is a HISO process or From Host Maintenance (FHM) process. No validation is performed.

Param Default No default value. If this parameter is not present in the LCONF, the processes that access the LCONF for this parameter abend during initialisation.

NBGC-BLOCKING-LOG-ILF

This Y/N (Yes/No) parameter indicates whether or not the NBGC Interchange Interface will log Online Blocking Messages (1324/1334) to the ILF.

Param Default N.

NBGC-EARLY-REVERSAL-TIME

This parameter indicates the time of day after which the NBGC Interchange Interface will stop looking in the previous day's ILF for matching 'early reversal' transactions.

Format of this param is HH:MM.

Param Default 24:00.

NBGC-STAN-PREFIX

A unique value used as the first digit in the generation of the Systems Audit Trace Number (STAN) used in external messages.

This mandatory parameter must be set up for each NBGC process running on the BASE24 system. The value of this parameter is 1 character in length in the format 0-9. All Issuer NBGC processes must have the value '0' and each Acquirer NBGC process must have a unique value in the range 1 – 9. **A maximum of 9 NBGC Interchange Interface acquirer processes can be configured on the system.**

Process Name Required (e.g. P1A^NBGC).

Param Default No default value. If this parameter is not present in the LCONF, the processes that access the LCONF for this parameter abend during initialisation.

PROBE-INTERVAL

Time interval between probes expressed in hundredths of a second. The valid range is 10 – 1500.

Param Default 1500.

SAF-NOTIFY-INTERVAL

The number of writes made to the Store And Forward File (SAF) between calculating the percentage of file full. Each time a write is made to the SAF, a counter is increased by one. When the counter is equal to the value specified in this parameter, a calculation is made to determine how full the SAF is. The counter is then reset to zero.

When the file reaches the percentage full specified in the SAF-NOTIFY-PERCENTAGE parameter, a message is logged.

This parameter can be set to any numeric value from 10 - 32 766.

Param Default 50.

SAF-NOTIFY-PERCENTAGE

The percentage of file full at which a message is logged indicating the SAF is becoming full. The percentage of file full is calculated at the interval indicated in the SAF-NOTIFY-INTERVAL parameter until more file space is made available.

This parameter can be set to any numeric value from 10 - 99.

Param Default 80.

SOCKET-PROBING

Flag that switches socket probing to the desired mode of operation. The valid values are ‘ON’ and ‘OFF’.

Param Default OFF.

SW-ILF-FILE-RETENTION

The number of days of ILFs, prior to the current business day’s ILF, which are to remain on the system. After the specified number of days, ILFs are purged from the system by the Interchange process.

For the NBGC Interchange Interface, ILFs are not used for reporting or extract purposes. It is therefore recommended that this should be set to a low number to conserve disk space on the system. The NBGC Interchange Interface requires the ILFs from yesterday and today to hold context for matching reversals to the original transaction. ILFs are not currently required for any other processing.

This param can be set to any numeric value from 3 - 99.

Param Default 7.

For parameters required for Hardware Security Module (HSM) use, refer to the ***BASE24 Transaction Security Manual***.

3.4 INTERCHANGE CONFIGURATION FILE (ICF)

The NBGC Interchange Interface process accesses data maintained in the Interchange Configuration File (ICF) to determine the variable parameters relevant to the IAN. These parameters define the processing of transactions between the NBGC Interchange Interface and the IAN.

For the NBGC Interchange Interface ICF record, data must be configured on the following screens:

- Screen 1
- Screen 2
- Screen 3
- Screen 7
- Screen 8
- Screen 12

The fields on these screens used by the NBGC Interchange Interface are described in the sections below.

Operator access to the ICF is permitted in accordance with the security access levels assigned in the BASE24 CRT access system. For more information on the BASE24 CRT access system, refer to the ***BASE24 CRT Access Manual.***

For more details on the ICF, refer to the ***BASE24 Base Files Maintenance Manual***.

3.4.1 ICF Screen 1

ICF screen 1 identifies the Interchange and the Interchange Interface process names and information. The fields displayed on the ICF screen 1 (Figure 3-1) are defined on the following pages.

```
BASE24-BASE  INTERCHANGE CONFIG        PBK3        99/07/20  15:54  01 OF 13
INTERCHANGE FIID:                         PROCESS:


                    SWITCH TYPE:
         INTERCHANGE LOGICAL NET:
                 REPORTING NAME:

                 INSTITUTION ID:
                      SWITCH ID:
                      STATION 1:
                      STATION 2:

                       SIC CODE: 0
                  CURRENCY CODE: 528  (NE)
               DEFAULT TERM NUM:
         DEFAULT ACQUIRER ID NUM: 00000000000
        CUSTOMER BALANCE DISPLAY: 0  (DON'T DISPLAY OR PRINT)

 RECORD LAST CHANGED: 99/02/03  18:16   BY USER: 0255 , 00000255   CHANGE
 *********************************** BASE24 **********************************
  NEW PAGE:        FILE DESTINATION:        NEW LOGICAL NETWORK ID:
                       F12-HELP
```

Figure 3-1 ICF Screen 1

INTERCHANGE FIID **4 (A) M**

This field is the institution identifier that uniquely identifies the Interchange. This field and the Process field form the primary key of the file.

PROCESS **16 (A) M**

This field is the name of the Interchange Interface process for the Interchange defined in NEF. The Process and Interchange FIID fields form the primary key of the file.

SWITCH TYPE **4 (A) M**

This field identifies the Interchange and allows access to the Interchange-specific screens. Enter a value of NBGC to allow the NBGC specific screen 12 to be displayed.

INTERCHANGE LOGICAL NET **4 (A) M**

This field is the name of the logical network of the Interchange Interface process. The name is defined in the NEF.

REPORTING NAME)

INSTITUTION ID)

SWITCH ID) Not used by the NBGC Interchange Interface.

STATION 1)

STATION 2)

SIC CODE)

CURRENCY CODE **3 (N) M**

This field identifies the currency code for used in transactions that the Interchange receives. For the NBGC Interchange Interface, a value of 978 (Euros) is specified.

Related Fields: IDF CURRENCY CODE and TDF CURRENCY CODE

DEFAULT TERM NUM)

DEFAULT ACQUIRER ID)
NUM) Not used by the NBGC Interchange Interface.

CUSTOMER BALANCE)
DISPLAY)

3.4.2 ICF Screen 2

The ICF screen 2 displays the Interchange settlement information. The fields displayed on the ICF screen 2 (Figure 3-2) are defined on the following pages.

```
BASE24-BASE  INTERCHANGE CONFIG       PBK3        99/07/20  15:56  02 OF 13
INTERCHANGE FIID:                        PROCESS:

                        SETTLEMENT INFORMATION
          SETTLEMENT HOUR: 23              SETTLEMENT MINUTE: 59
          SETTLEMENT DAYS: 1  (PROCESS 7 DAYS)
          REPORT PRIORITY: 100                    REPORT CPU:  0

                          SWITCH POSTING DATE (YYMMDD)


                             HOLIDAY DATES   (YYMMDD)



                         REPORT DATA MASKING PARAMETERS
       DATA MASK FLAG: Y  (MASK SENSITIVE DATA)      RIGHT UNMASKED DIGITS: 4
    MIN MASKED DIGITS: 9                          MAX LEFT UNMASKED DIGITS: 0

 RECORD LAST CHANGED: 99/02/03  18:16   BY USER: 0255 , 00000255   CHANGE
 *********************************** BASE24 **********************************
  NEW PAGE:        FILE DESTINATION:        NEW LOGICAL NETWORK ID:
                        F12-HELP
```

Figure 3-2 ICF Screen 2

SETTLEMENT INFORMATION

The following fields contain Interchange settlement information.

SETTLEMENT HOUR **2 (N) M**

The hour (local time) at which settlement for this interchange occurs. Valid values are 00 to 23.

Default Value: 00

SETTLEMENT MINUTE **2 (N) M**

The minute (local time) at which settlement for this interchange occurs. Valid values are 00 to 59.

Default Value: 00

SETTLEMENT DAYS **1 (N) M**

This field indicates the processing schedule for the Interchange processes and is set to 1 (process 7 days).

Default Value: 1

REPORT PRIORITY

This field is not used by the NBGC Interchange Interface.

REPORT CPU

This field is not used by the NBGC Interchange Interface.

SWITCH POSTING DATE (YYMMDD) **6 (N) Y**

This field displays the current NBGC Interchange posting date. The Interchange Interface process will update this field at initialisation and Interchange cutover.

HOLIDAY DATES (YYMMDD)

This field is not used by the NBGC Interchange Interface.

REPORT DATA MASKING PARAMETERS

The values in the following fields specify the report data masking parameters. These parameters support the masking of sensitive information in accordance with PCI data security standards.

DATA MASK FLAG **1 (A) M**

A code indicating whether sensitive data should be masked or unmasked. Valid values are as follows:

Y = Mask sensitive data
N = Do not mask sensitive data.

Default Value: Y

RIGHT UNMASKED DIGITS **1 (N) M**

A code defining the number of rightmost digits to be displayed unmasked. Valid values are as follows:

0–9 = Number of rightmost digits to be displayed unmasked.

Default Value: 4

MIN MASKED DIGITS **1 (N) M**

A code defining the minimum number of digits to be masked. Valid values are as follows:

0–9 = Minimum number of digits to be masked.

Default Value: 9

MAX LEFT UNMASKED DIGITS **1 (N) M**

A code defining the maximum number of leftmost digits to be displayed unmasked. Valid values are as follows:

0–9 = Maximum number of leftmost digits to be displayed unmasked.

Default Value: 0

3.4.3 ICF Screen 3

The ICF screen 3 defines the timers and processing parameters. The fields displayed on the ICF screen 3 (Figure 3-3) are defined on the following pages.

```
BASE24-BASE  INTERCHANGE CONFIG       PBK3        99/07/20  15:57  03 OF 13
INTERCHANGE FIID:                        PROCESS:

         TIMER LIMITS                    PROCESSING OPTIONS

   NETWORK MANAGEMENT:   30 (SEC)                ACQUIRER: N (Y/N)
     EXTENDED NETWORK:   60 (SEC)                  ISSUER: N (Y/N)
     WAIT FOR TRAFFIC:   60 (SEC)       PROCESSING MODE:
   PERFORMANCE PERIOD:   20 (MIN)     AUTO SIGNON START: N (Y/N)
                                       MAXIMUM TIMEOUTS:    2
 MAXIMUM OUTSTANDING TRANSACTIONS         MAX SAF RETRY:    0
      OUTBOUND:   15                      ACK TO SWITCH: N (Y/N)
       INBOUND:   15                    ACK FROM SWITCH: N (Y/N)


    NETWORK MANAGEMENT MESSAGE ENABLED: N  (Y = ENABLED, N = DISABLED)
           TYPE OF INTERCHANGE REPORTS: 0  (ATM/DETAIL AND SETTLEMENT)
                    ILF EXTRACT NUMBER: 3

 RECORD LAST CHANGED: 99/02/03  18:16   BY USER: 0255 , 00000255   CHANGE
 ********************************** BASE24 **********************************
  NEW PAGE:        FILE DESTINATION:        NEW LOGICAL NETWORK ID:
                       F12-HELP
```

Figure 3-3 ICF Screen 3

TIMER LIMITS

The following timer fields are used to control specific transaction message activity between the BASE24-atm network and the Interchange. The values entered in these fields must be checked against the Interchange timer values to ensure the integrity of message traffic.

NETWORK MANAGEMENT **4 (N) M**

The maximum number of seconds that BASE24-atm waits for a response to a network management message such as logon requests and echo tests.

Default Value: 30 (SEC)

EXTENDED NETWORK **4 (N) M**

The maximum number of seconds that BASE24-atm waits before attempting to retry a network management message during periods when the line to the Interchange is down.

Default Value: 60 (SEC)

WAIT FOR TRAFFIC **4 (N) M**

The maximum number of seconds that BASE24-atm waits for traffic on the line from the Interchange before initiating a network management message.

Default Value: 60 (SEC)

PERFORMANCE PERIOD **4 (N) M**

This field displays the length of time (in minutes) of an interchange performance monitoring session.

Default Value: 20 (MIN)

PROCESSING OPTIONS

The following fields define the processing options for the Interchange.

ACQUIRER)

ISSUER) Not used by the NBGC Interchange Interface.

PROCESSING MODE)

AUTO SIGNON START **1 (A) M**

This field specifies whether the Interchange Interface process is set to automatically sign-on at start up or logon to the Interchange without operator intervention. It is recommended that for normal operation, this field is set to Y. The valid values are:

N start up is not set to sign-on or logon to the Interchange automatically

Y start up is set to sign-on or logon to the Interchange automatically

MAXIMUM TIMEOUTS **4 (N) M**

The maximum number of consecutive time-outs allowed before network management measures are taken to determine the status of the link.

MAXIMUM OUTSTANDING TRANSACTIONS

The Maximum Outstanding Transactions consists of two fields:

- Outbound
- Inbound

OUTBOUND **4 (N) M**

The maximum number of outgoing transactions that can be waiting to be processed. The recommended value is 15.

INBOUND **4 (N) M**

The maximum number of incoming transactions that can be waiting to be processed. The recommended value is 15.

MAXIMUM SAF RETRY **4 (N) M**

The maximum number of times a transaction from the SAF attempts to be transmitted to the Interchange before the record is written to a hard-copy log and deleted from the file. A value of '0' implies retry indefinitely.

ACK TO SWITCH)

ACK FROM SWITCH)

NETWORK MANAGEMENT) Not used by the NBGC Interchange Interface.

MESSAGE ENABLED)

TYPE OF INTERCHANGE)
REPORTS)

ILF EXTRACT NUMBER)

3.4.4 ICF Screen 7

The ICF screen 7 defines the BASE24-atm allowable transactions. The fields displayed on the ICF screen 7 (Figure 3-4) are defined on the following pages.

```
BASE24-ATM   INTERCHANGE CONFIG       PBK3        99/07/20  15:57  07 OF 13
INTERCHANGE FIID:                        PROCESS:

                            ATM ICF DATA
          NOT ON US                      TRANSACTIONS ALLOWED TO SWITCH
      WITHDRAWAL           0              WITHDRAWAL           N
      WITHDRAWAL CC        0              WITHDRAWAL CC        N
      DEPOSIT              0              DEPOSIT              N
      INQUIRY              0              INQUIRY              N
      TRANSFER             0              TRANSFER             N
      ELECTRONIC PAYMENT   0              ELECTRONIC PAYMENT   N
      PAYMENT ENCLOSED     0              PAYMENT ENCLOSED     N
      CASH CHECK           0              CASH CHECK           N
      MSG TO INSTITUTION   0              MSG TO INSTITUTION   N
      PIN CHANGE           0              PIN CHANGE           N
      SPLIT DEPOSIT        0              SPLIT DEPOSIT        N

VALUES FOR NOT ON US ARE '0' THRU '4', TRANS ALLOWED TO SWITCH ARE 'Y' OR 'N'
0 = NOT ALLOWED  1 = INTRACOUNTY  2 = INTRASTATE  3 = INTERSTATE  4 = INTNAT'L
 RECORD LAST CHANGED: 99/02/03  18:16   BY USER: 0255 , 00000255   CHANGE
 ********************************** BASE24 **********************************
  NEW PAGE:        FILE DESTINATION:        NEW LOGICAL NETWORK ID:
                        F12-HELP
```

Figure 3-4 ICF Screen 7

ATM ICF DATA

The following fields define the BASE24-atm allowable transactions.

NOT ON US **1 (N) M**

This field defines which transaction types listed under the field are allowed or disallowed, where BASE24-atm cardholders use ATMs connected to and controlled by institutions serviced by an interchange. For each type of transaction listed, the valid values are:

0 Not allowed (Default)

1 Allowed within county

2 Allowed within state

3 Allowed nationally

4 Allowed internationally

For the NBGC Interchange only withdrawal and inquiry transactions are allowed. A value of 4 should be entered alongside these transactions; all others should be set to 0.

TRANSACTIONS ALLOWED TO SWITCH **1 (A) Y**

This field defines the transaction types listed under the field are allowed or disallowed, where the Interchange cardholders use ATMs for which BASE24 is acting as the acquirer. For each type of transaction listed, the valid values are:

N Not allowed (Default)

Y Allowed

For the NBGC Interchange only withdrawal and inquiry transactions are allowed. A value of Y should be entered alongside these transactions; all others should be set to N.

3.4.5 ICF Screen 8

The ICF screen 8 defines the BASE24-atm sharing groups and the business traffic timers. The fields displayed on the ICF screen 8 (Figure 3-5) are defined on the following pages.

```
BASE24-ATM   INTERCHANGE CONFIG       PBK3        99/07/20  15:57  08 OF 13
INTERCHANGE FIID:                        PROCESS:


                                 ATM ICF DATA

                   AUTH PROCESS:
          DEFAULT ROUTING GROUP: 00000000000

                                 TIMER LIMITS
             STORE AND FORWARD:   30 (SEC)
                      OUTBOUND:   15 (SEC)
                       INBOUND:   15 (SEC)
                    COMPLETION:   60 (SEC)
                COMPLETION ACK:   30 (SEC)

 SHARING GROUPS:


 RECORD LAST CHANGED: 99/02/03  18:16   BY USER: 0255 , 00000255   CHANGE
 ********************************** BASE24 **********************************
  NEW PAGE:        FILE DESTINATION:        NEW LOGICAL NETWORK ID:
                       F12-HELP
```

Figure 3-5 ICF Screen 8

ATM ICF DATA

AUTH PROCESS **16 (A) M**

This field defines the authorisation process to which the Interchange routes ATM transactions received from the Switch.

DEFAULT ROUTING GROUP

This field is not used by the NBGC Interchange Interface.

TIMER LIMITS

STORE AND FORWARD **4 (N) M**

This field specifies the time limit associated with the store-and-forward response from an interchange for an ATM transaction. The recommended time limit is 60 seconds to ensure that a reversal repeat message is not re-transmitted more frequently than once a minute.

Default Value: 30 (SEC)

OUTBOUND **4 (N) M**

This field specifies the time limit associated with transaction responses from the Interchange for ATM transactions. The recommended value is 30 seconds. Note that the value should be less than the time-out value used by the ATMs.

Default Value: 30 (SEC)

INBOUND **4 (N) M**

This field specifies the time limit associated with transaction responses from BASE24-atm.

Default Value: 30 (SEC)

COMPLETION

This field is not used by the NBGC Interchange Interface.

COMPLETION ACK

This field is not used by the NBGC Interchange Interface.

SHARING GROUPS **1 (A) M**

A maximum of 24 fields are allowed. Each field specifies a one-character code to identify interchange sharing groups for both incoming and outgoing transactions. Zero is not a valid entry. Codes are not separated by blank spaces; however, unused portions to the right remain blank. Note that the sharing group(s) entered here must correspond to one or more of the sharing groups entered in the BASE24 TDF.

Related Field: TDF SHARING GROUPS

3.4.6 ICF Screen 12

The ICF screen 12 displays the BASE24 processing information specific to the NBGC Interchange Interface. The fields displayed on the ICF screen 12 (Figure 3-6) are defined on the following pages.

```
BASE24-ATM   ICF FILE MAINTENANCE     PBK3        99/07/20  15:47  12 OF 13
INTERCHANGE FIID:                        PROCESS:

                                   NBGC DATA

   STATION NAMES:     ________________     ________________
                      ________________     ________________



   TRANSPORT KEY EXCHANGE:                          RETRY LIMITS
     EXCHANGE KEYS:             _ __________          MAX BAD MESSAGES:   ____
     DAILY KEY EXCHANGE TIME:   ____ (hhmm)           MAX BAD MACS:       ____
     MAX SESSIONS:              ___
     SESSIONS USED:             ___




********************************** BASE24 ***********************************
  NEW PAGE:        FILE DESTINATION:        NEW LOGICAL NETWORK ID:
                               F12-HELP
```

Figure 3-6 ICF Screen 12

NBGC DATA

The following fields define NBGC specific data.

STATION NAMES **16 (A) M**

The BASE24 station names, as declared in the NEF, used for communication with the IAN.

TRANSPORT KEY EXCHANGE

EXCHANGE KEYS **1 (A) M**

A (Yes/No) indicator specifying whether the transport key exchange is enabled or disabled. Valid values are:

Y Key Exchange is Enabled

N Key Exchange is Disabled

If this field is set to 'Y' (Key Exchange Enabled), The Daily Key Exchange Time field must contain a valid time.

DAILY KEY EXCHANGE TIME **4 (N) M**

This field displays the time at which the session key is exchanged by means of an enforced LOGOFF/LOGON sequence. For more information see section 3.5.

MAX SESSIONS **3 (N) M**

This field specifies the maximum number of sessions that will take place, and session keys derived, before a new transport key is generated and exchanged. This field also specifies the maximum number of times that the interface can retry a logon before operator intervention is required.

SESSIONS USED **3 (N) M**

This field specifies the number of sessions that have taken place under the current transport key. This field is also incremented whenever the IAN rejects a logon advice.

RETRY LIMITS

The following maximum retry limits are used to control NBGC specific functions.

MAX BAD MESSAGES **1 (N) M**

This field specifies the number of consecutive erroneous messages that an Interchange process can receive before the Interchange is automatically signed-off pending operator intervention.

Default Value: 2

MAX BAD MACS **1 (N) M**

This field specifies the number of consecutive MAC verify failures that an Interchange process can receive before the Interchange is automatically signed-off pending operator intervention.

If this field is set to a value greater than 1, an automatic key exchange is performed with the IAN when the count of MAC verify failures reaches 1.

Default Value: 1

3.5 KEY FILE (KEYF)

The KEYF contains the information and parameters required by the BASE24 Interface processes for PIN encryption, PIN translation, message authentication and dynamic key management.

If the BASE24 system uses the Transaction Security Services (TSS), then the key locators held on the KEYF reference the actual security keys held on the TSS database.

The ISO Host Interface process and Interchange Interface processes use the KEYF instead of the KEY6 when the LCONF param DES-TRIPLE-SINGLE is either set to a value of N or not present. Otherwise these processes use the KEY6.

The KEYF contains one record for each:

- DPC/Host Interface process
- Interchange Interface process

The key to records in the KEYF is a combination of the values in the DPC/FIID and INTERFACE PROCESS fields.

The following screens are used to access data in the KEYF:

- **Screen 1** contains PIN and MAC processing parameters, key locators to intermediate keys, and key locators to PIN and MAC key exchange keys.
- **Screen 2** contains key locators to inbound and outbound keys for PINs and MACs.
- **Screen 3** contains dynamic key management parameters.
- **Screen 4** contains message encryption parameters.

The fields on these screens used by the NBGC Interchange Interface are described in the sections below.

Screen 2 is not covered here, as the NBGC Interchange does not support the user configuration of PIN session keys and MAC session keys.

Screen 4 is not covered here, as the NBGC Interchange does not support message encryption.

For more information on the usage of the KEYF, refer to the following:

- ***BASE24 Base Files Maintenance Manual***
- ***BASE24 Transaction Security Manual***

3.5.1 KEYF Screen 1

The KEYF screen 1 contains PIN encryption, PIN translation, message authentication parameters, intermediate keys, PIN key exchange keys and Message Authentication Code (MAC) key exchange keys. The fields displayed on the KEYF screen 1 (Figure 3-7) are defined on the following pages.

```
 BASE24-BASE  KEY FILE                PBK3        99/07/20  17:06  01 OF 04
         DPC/FIID:               INTERFACE PROCESS:

    ENCRYPT TYPE: _ ________________   BASE24 ENCRYPT TYPE: _ _______________
PIN BLOCK FORMAT: _ _______________        ANSI PAN FORMAT: _ _______________
MAC ENCRYPT TYPE: _ ____________________ PIN PAD CHARACTER: _
   MAC DATA TYPE: _ ________                NUMBER OF KEYS: _ __________
      KEY LENGTH: _ ________              FULL MESSAGE MAC: _ _______________

                                INTERMEDIATE KEYS

          CLEAR: ________________                    CHECK DIGITS: ____
      ENCRYPTED: ________________

                                  EXCHANGE KEYS

        PIN KEY: ________________  ________________  CHECK DIGITS: ____
        MAC KEY: ________________  ________________  CHECK DIGITS: ____

*********************************** BASE24 **********************************
 NEW PAGE:        FILE DESTINATION:        NEW LOGICAL NETWORK ID:
                        F12-HELP
```

Figure 3-7 KEYF Screen 1

DPC/FIID **4 (AN) M**

The FIID of the interchange using this KEYF record. The value in this field matches the INTERCHANGE FIID field on the ICF screen 1.

Default Value: No default value.

INTERFACE PROCESS **16 (AN) M**

The name of the interface process associated with the interchange, identified in the DPC/FIID field. The value in this field matches the value in the PROCESS field on the ICF screen 1.

Default Value: No default value.

ENCRYPT TYPE **1 (N) M**

This field specifies the type of PIN encryption used or expected by the DPC or interchange. The valid values are:

0 Clear PINs (Default)

1 Security module PIN encryption

2 Software DES PIN encryption

This field must be set to 1 for the NBGC Interchange Interface.

BASE24 ENCRYPT TYPE **1 (N) M**

This field specifies the type of PIN management used by BASE24. The valid values are:

0 Clear PINs (Default)

1 Security module PIN management

This field must be set to 1 for the NBGC Interchange Interface.

PIN BLOCK FORMAT **1 (N) M**

This field specifies the PIN block format of the PIN in the inbound and outbound external messages. The valid values are:

0 Clear PINs (Default)

1 ANSI PIN block (PIN/PAN PIN block)

3 PIN/PAD PIN block

Option 1 (ANSI PIN block) is also known as the PIN/PAN PIN block because it includes three PAN formats, only one of which is part of the ANSI standard. The PAN format being used is specified in the ANSI PAN FORMAT field on this screen.

This field must be set to 1 for the NBGC Interchange Interface.

ANSI PAN FORMAT **1 (N) M**

If the value in the PIN BLOCK FORMAT field is 1, as required for the NBGC Interchange Interface, the value in this field specifies which PAN digits are used in the formation of the external message PIN/PAN PIN block. The valid values are:

0 12 right-most digits of the PAN excluding the check digit (ANSI standard) (Default)

1 12 right-most digits of the PAN including the check digit

2 12 left-most digits of the PAN

The ANSI standard is one of three PAN formats available with PIN/PAN PIN blocks. While this field is called the ANSI PAN FORMAT, it also includes two other PAN formats that are not part of the ANSI standard.

This field must be set to 0 for the NBGC Interchange Interface.

MAC ENCRYPT TYPE **1 (N) M**

This field specifies how message authentication will be performed for messages between the DPC or interchange and the interface.

Default Value: 0

This field must be set to 1 for the NBGC Interchange Interface.

PIN PAD CHARACTER **1 (A) M**

If the value in the PIN BLOCK FORMAT field is 3 (PIN/PAD), the value in this field specifies the PAD character used in the formation of the external message's PIN block. The valid values are A - F.

Default Value: 0

This field must be set to 'F' for the NBGC Interchange Interface.

MAC DATA TYPE **1 (N)**

A code specifying the character set in which messages are being transmitted between the DPC or interchange and the interface. The valid values are 0 (ASCII) and 1 (EBCDIC).

Default Value: 0

This field must be set to '0' for the NBGC Interchange Interface.

NUMBER OF KEYS **1 (N) M**

A code identifying whether the inbound and outbound keys are combined or separate. This field is not used by the NBGC Interchange Interface and should be set to the default setting of '1' (Combined).

KEY LENGTH **1 (N) M**

A code identifying whether single-length or double-length Key Exchange Keys (KEKs) are being used with this interface. This field is not used by the NBGC Interchange Interface and should be set to the default setting of '1' (Single-length KEK).

FULL MESSAGE MAC **1 (A) M**

A code specifying whether selected data elements or the entire message will be considered when computing the MAC. This field is not used by the NBGC Interchange Interface and should be set to the default setting of 'N' (Selected fields authenticated).

INTERMEDIATE KEYS

Intermediate Keys are utilised whenever a PIN that has been encrypted under a secure key (security module PIN encryption) must be decrypted into the clear or when a PIN is in the clear and needs to be encrypted under a secure key.

These fields are not used by the NBGC Interchange Interface.

EXCHANGE KEYS

When PIN or MAC keys are automatically exchanged between the Interchange Interface and the interchange, they are to be encrypted under this key for the interchange. PIN or MAC keys that an Interchange Interface receives from an interchange are also encrypted under this key. The exchange key must be manually exchanged between the two entities.

PIN KEY **32 hexadecimal characters M**

The key locator to the security module encrypted form of the double length key (BGC PIN MASTER KEY) used to exchange PIN KEYS in the TSS database. Valid values for each character in the field are 0-9 and A-F. Only the first 16 characters in this field are used as the key locator.

Default Value: 00000000000000000000000000000000

CHECK DIGITS **4 hexadecimal characters M**

This field is not used by the NBGC Interchange Interface.

MAC KEY **32 hexadecimal characters M**

The key locator to the security module encrypted form of the double length key (BGC MAC MASTER KEY) used to exchange MAC KEYS in the TSS database. Valid values for each character in the field are 0-9 and A-F. Only the first 16 characters in this field are used as the key locator.

Default Value: 00000000000000000000000000000000

CHECK DIGITS **4 hexadecimal characters M**

This field is not used by the NBGC Interchange Interface.

3.5.2 KEYF Screen 3

The KEYF screen 3 contains dynamic key management parameters. The fields displayed on KEYF screen 3 are not shown, because the NBGC Interchange Interface uses only one of the fields:

KEY PROCESSING TYPE **1 (A) M**

A code identifying the type of dynamic key management processing this interface can perform. The valid values are:

C Co-network. Each interface is responsible for generating its outbound key and can request its inbound key from the other interface in the co-network.

M Main. This interface is responsible for all key generation.

N None. This interface does not generate or receive keys.

S Secondary. This interface can request the main process to generate keys, but does not generate keys itself.

A description of the code entered is displayed to the right of the KEY PROCESSING TYPE field.

Default Value: N

If there is a requirement to inhibit the generation of logon messages by the NBGC Interchange Interface (i.e. the process is used to emulate the IAN), then this field must be set to ‘S’.

The other fields on KEYF Screen 3 are not used, as the NBGC Interchange Interface does not require the user configuration of dynamic key management parameters.

3.6 KEY FILE (KEY6)

The Key File (KEY6) contains the information and parameters required by release 6.0 BASE24 Host Interface and Interchange Interface processes running in release 5.x or 6.0 networks for PIN encryption, PIN translation, message authentication and dynamic key management.

The KEY6 allows for double length encryption keys which are required when translating PINS from encryption under a double length key using the Triple DES algorithm to encryption under a single length key using the Single DES algorithm and vice versa.

The ISO Host Interface process and Interchange Interface processes use the KEY6 instead of the KEYF when the LCONF param DES-TRIPLE-SINGLE is set to a value of Y. Otherwise these processes use the KEYF.

The KEY6 contains one record for each:

- DPC/Host Interface process
- Interchange Interface process

The following screens are used to access records in the KEY6:

- **Screen 1** contains PIN encryption, PIN translation, and message authentication parameters, intermediate keys, PIN key exchange keys, and message authentication code (MAC) key exchange keys.
- **Screen 2** contains inbound and outbound keys for PINs.
- **Screen 3** contains inbound and outbound keys for MACs.
- **Screen 4** contains dynamic key management parameters.

The fields on these screens used by the NBGC Interchange Interface are described in the sections below.

Screen 2 is not covered here, as the NBGC Interchange does not support the user configuration of PIN session keys.

Screen 3 is not covered here, as the NBGC Interchange does not support the user configuration of MAC session keys.

For more information on the usage of the KEY6, refer to the following:

- ***BASE24 Base Files Maintenance Manual***
- ***BASE24 Transaction Security Manual***

3.6.1 KEY6 Screen 1

The KEY6 screen 1 contains PIN encryption, PIN translation, message authentication parameters, intermediate keys, PIN key exchange keys and Message Authentication Code (MAC) key exchange keys. The fields displayed on the KEY6 screen 1 (Figure 3-8) are defined on the following pages.

```
 BASE24-BASE  KEY6 FILE                PBK3        99/07/20  17:06  01 OF 04
        DPC/FIID:               INTERFACE PROCESS:

    ENCRYPT TYPE: _ ________________   BASE24 ENCRYPT TYPE: _ _______________
PIN BLOCK FORMAT: _ _______________        ANSI PAN FORMAT: _ _______________
MAC ENCRYPT TYPE: _ ____________________ PIN PAD CHARACTER: _
   MAC DATA TYPE: _ ________               NUMBER OF KEYS: _ __________
      KEY LENGTH: _ ________             FULL MESSAGE MAC: _ _______________
  MAC KEY LENGTH: _ ________
                               INTERMEDIATE KEYS

          CLEAR: ________________                    CHECK DIGITS: ____
      ENCRYPTED: ________________

                                 EXCHANGE KEYS

        PIN KEY: ________________  ________________  CHECK DIGITS: ____
        MAC KEY: ________________  ________________  CHECK DIGITS: ____

*********************************** BASE24 **********************************
 NEW PAGE:        FILE DESTINATION:       NEW LOGICAL NETWORK ID:
                       F12-HELP
```

Figure 3-8 KEY6 Screen 1

DPC/FIID **4 (AN) M**

The FIID of the interchange using this KEY6 record. The value in this field matches the INTERCHANGE FIID field on the ICF screen 1.

Default Value: No default value.

INTERFACE PROCESS **16 (AN) M**

The name of the interface process associated with the interchange, identified in the DPC/FIID field. The value in this field matches the value in the PROCESS field on the ICF screen 1.

Default Value: No default value.

ENCRYPT TYPE **1 (N) M**

This field specifies the type of PIN encryption used or expected by the DPC or interchange. The valid values are:

0 Clear PINs (Default)

1 Security module PIN encryption

2 Software DES PIN encryption

This field must be set to 1 for the NBGC Interchange Interface.

BASE24 ENCRYPT TYPE **1 (N) M**

This field specifies the type of PIN management used by BASE24. The valid values are:

0 Clear PINs (Default)

1 Security module PIN management

This field must be set to 1 for the NBGC Interchange Interface.

PIN BLOCK FORMAT **1 (N) M**

This field specifies the PIN block format of the PIN in the inbound and outbound external messages. The valid values are:

0 Clear PINs (Default)

1 ANSI PIN block (PIN/PAN PIN block)

3 PIN/PAD PIN block

Option 1 (ANSI PIN block) is also known as the PIN/PAN PIN block because it includes three PAN formats, only one of which is part of the ANSI standard. The PAN format being used is specified in the ANSI PAN FORMAT field on this screen.

This field must be set to 1 for the NBGC Interchange Interface.

ANSI PAN FORMAT **1 (N) M**

If the value in the PIN BLOCK FORMAT field is 1, as required for the NBGC Interchange Interface, the value in this field specifies which PAN digits are used in the formation of the external message PIN/PAN PIN block. The valid values are:

0 12 right-most digits of the PAN excluding the check digit (ANSI standard) (Default)

1 12 right-most digits of the PAN including the check digit

2 12 left-most digits of the PAN

The ANSI standard is one of three PAN formats available with PIN/PAN PIN blocks. While this field is called the ANSI PAN FORMAT, it also includes two other PAN formats that are not part of the ANSI standard.

This field must be set to 0 for the NBGC Interchange Interface.

MAC ENCRYPT TYPE **1 (N) M**

This field specifies how message authentication will be performed for messages between the DPC or interchange and the interface.

Default Value: 0

This field must be set to 1 for the NBGC Interchange Interface.

PIN PAD CHARACTER **1 (A) M**

If the value in the PIN BLOCK FORMAT field is 3 (PIN/PAD), the value in this field specifies the PAD character used in the formation of the external message's PIN block. The valid values are A - F.

Default Value: 0

This field must be set to F for the NBGC Interchange Interface.

MAC DATA TYPE **1 (N)**

A code specifying the character set in which messages are being transmitted between the DPC or interchange and the interface. The valid values are 0 (ASCII) and 1 (EBCDIC).

Default Value: 0

This field must be set to 0 for the NBGC Interchange Interface.

NUMBER OF KEYS **1 (N) M**

A code identifying whether the inbound and outbound keys are combined or separate. This field is not used by the NBGC Interchange Interface and should be set to the default setting of '1' (Combined).

KEY LENGTH **1 (N) M**

A code identifying whether single-length or double-length PIN Key Exchange Keys (KEKs) are being used with this interface. This field is not used by the NBGC Interchange Interface and should be set to the default setting of '1' (Single-length KEK).

FULL MESSAGE MAC **1 (A) M**

A code specifying whether selected data elements or the entire message will be considered when computing the MAC. This field is not used by the NBGC Interchange Interface and should be set to the default setting of 'N' (Selected fields authenticated).

MAC KEY LENGTH **1 (N) M**

A code identifying whether single-length or double-length MAC Key Exchange Keys (KEKs) are being used with this interface. This field is not used by the NBGC Interchange Interface and should be set to the default setting of '1' (Single-length KEK).

INTERMEDIATE KEYS

Intermediate Keys are utilised whenever a PIN that has been encrypted under a secure key (security module PIN encryption) must be decrypted into the clear or when a PIN is in the clear and needs to be encrypted under a secure key.

CLEAR **16 hexadecimal characters**

This field is not used by the NBGC Interchange Interface.

CHECK DIGITS **4 hexadecimal characters**

This field is not used by the NBGC Interchange Interface.

ENCRYPTED **16 hexadecimal characters**

Security module encrypted intermediate key. This key is used for the interim Triple DES solution when a PIN block encrypted using a double-length key (Triple DES) is received from an interchange. The PIN block is translated to encryption under the intermediate key (Single DES) by the security module.

EXCHANGE KEYS

When PIN or MAC keys are automatically exchanged between the Interchange Interface and the interchange, they are to be encrypted under this key for the interchange. PIN or MAC keys that an Interchange Interface receives from an interchange are also encrypted under this key. The exchange key must be manually exchanged between the two entities.

PIN KEY **32 hexadecimal characters M**

The security module encrypted form of the double length key (BGC PIN MASTER KEY) used to exchange PIN KEYS. Valid values for each character in the field are 0-9 and A-F.

Default Value: 00000000000000000000000000000000

CHECK DIGITS **4 hexadecimal characters M**

This field is not used by the NBGC Interchange Interface.

MAC KEY **32 hexadecimal characters M**

The security module encrypted form of the double length key (BGC MAC MASTER KEY) used to exchange MAC KEYS. Valid values for each character in the field are 0-9 and A-F.

Default Value: 00000000000000000000000000000000

CHECK DIGITS **4 hexadecimal characters M**

This field is not used by the NBGC Interchange Interface.

3.6.2 KEY6 Screen 4

The KEY6 screen 4 contains dynamic key management parameters. The fields displayed on KEY6 screen 4 are not shown, because the NBGC Interchange Interface uses only one of the fields:

KEY PROCESSING TYPE **1 (A) M**

A code identifying the type of dynamic key management processing this interface can perform. The valid values are:

C Co-network. Each interface is responsible for generating its outbound key and can request its inbound key from the other interface in the co-network.

M Main. This interface is responsible for all key generation.

N None. This interface does not generate or receive keys.

S Secondary. This interface can request the main process to generate keys, but does not generate keys itself.

A description of the code entered is displayed to the right of the KEY PROCESSING TYPE field.

Default Value: N

If there is a requirement to inhibit the generation of logon messages by the NBGC Interchange Interface (i.e. the process is used to emulate the IAN), then this field must be set to ‘S’.

The other fields on KEY6 Screen 4 are not used, as the NBGC Interchange Interface does not require the user configuration of dynamic key management parameters.

3.7 LOG MESSAGE CONFIGURATION FILE (LMCF)

The LMCF contains a record for each modified log message in the system. The NBGC Interchange Interface process opens the LMCF, retrieves its modified log messages, and closes the LMCF. Whenever the NBGC Interchange Interface process generates a log message, it checks the LMCF records in memory to determine whether the message requires modification.

3.8 TOKEN FILE (TKN)

To log or extract tokens, the Token File (TKN) must be configured appropriately. The token parameters can be set on the TKN file maintenance screen.

During initialisation, the NBGC Interchange Interface reads the TKN file for the records that specify which tokens should be logged to the ILF.

The TOKEN GROUP key field is filled from the Interchange FIID field (on ICF Screen 1). If the TKN records are found, then they are stored in memory. If the TKN records are not found, then all tokens are logged to the ILF.

The following global product tokens used by the NBGC Interchange Interface can be logged or extracted:

- 03 - ATM 5.0 token
- 25 - Surcharge Data token
- B0 - Acquirer Switch token
- B1 - Issuer Switch token
- BE - Original Currency token
- BH - Reversal Date and Time token

For information on the NBGC definition of the generic switch token, refer to Appendix B.

The following country product tokens used by the NBGC Interchange Interface can be logged or extracted:

- X0 - NBGC Transaction token
- X1 - NBGC Reversal token

The NBGC Interchange Interface does not currently use the other Dutch-specific country product BASE24 token, called the NBGC PAN token (Token Id = X2).

For information on the Dutch-specific token definitions, refer to Appendix B.

Note that no tokens are logged to the ILF when an “early” reversal message is received from the IAN.

3.9 EXTERNAL MESSAGE FILE (EMF)

External Message File (EMF) records specify which data elements are to be included in an external message for incoming and outgoing messages.

All BIM messages have default bitmaps specified by the NBGC Interchange Interface module, which correspond to the default bitmaps specified by the BIM protocol. The EMF maintains records for the NBGC Interchange Interface process that can be used to override the standard default bitmaps.

EMF functionality is only partially implemented for NBGC Interchange Interface processing, with only the following message types supported:

- 0200;
- 0210;
- 0420;
- 0430.

A value of ‘NBGC’ is supported as a valid Interface Type on EMF Screen 1. The NBGC Interchange Interface process will only attempt to read records where the INTERF TYP field contains ‘NBGC’, the DPC/MOD # field contains zero, the PROD # field contains ‘01’ (ATM), and the MSG TYP field contains one of the values listed above.

3.10 OTHER BASE24 FILES

Certain fields in other BASE24 files must be set to appropriate values to ensure that the required data is sent in messages to the IAN.

3.10.1 Terminal Data File (ATD/TDF)

Certain data elements that are stored in the Terminal Data File (ATD/TDF) are used to supply data required in the BIM messages that are sent to the IAN. All ATD/TDF records should, therefore, be set up to contain the correct data according to the rules of the BIM protocol. The fields that are used in the construction of switch messages are listed below.

For more information about the ATD/TDF, refer to the ***BASE24-atm Files Maintenance Manual***.

INST-ID-NUM **(INST ID NUM on the ATD screen)**

This field specifies the P-32 Acquirer Institution Identifier. All ATD/TDF records must contain a valid Acquirer Institution ID for the financial institution. The range of valid Acquirer Institution IDs for each financial institution can be supplied by NBGC.

TERM-ID **(TERM ID on the ATD screen)**

This field specifies the P-41 Card Acceptor Term Identifier which is six characters known to NBGC.

TERM-OWNER-NAME **(OWNER NAME on the ATD screen)**

TERM-CITY **(CITY on the ATD screen)**

POSTAL-CDE **(POSTAL CODE on the ATD screen)**

TERM-OWNER-NAME, TERM-CITY and POSTAL-CDE are used to format the first 38 characters of the P-43 Card Acceptor Name Location in the format: NAME>CITY>POST CODE.

CURRENCY-CDE **(CURRENCY on the ATD screen)**

This field specifies the type of currency and is used by the P-49 Currency Code Transaction. All ATD/TDF records must be set up with a currency code of 978 (Euros).

TERM-SUR-PROFILE (SURCHARGE PROFILE on the ATD screen)

This field is used for acquirers supporting ATM surcharging in the NBGC Interchange Interface. For non-bank ATMs, the first two characters of this field must match the required Device Type in Format 07 of P-61 in the BIM request. Any value not matching a non-bank Device Type (currently ‘02’ and ‘03’) will be mapped to ‘01’ (indicating a bank ATM).

In addition, the SHARING GROUP IDs are set up to correspond with the sharing groups specified for the IDF and ICF.

3.10.2 Surcharge File (SURF)

The Surcharge File (SURF) contains one record for each card group targeted by a terminal group for transaction acquirer fees in a particular currency.

Card groups are separated according to transaction routing method, e.g. either CPF or SPROUTE. Basic configuration of surcharges is driven by the relationship between terminal groups (i.e. ATD Surcharge Profile IDs) and card groups (i.e. Issuers). Within this relationship, specific card types, transaction types, account types and surcharge calculation methods are used to determine the surcharge amount.

For more information about the SURF, refer to the ***BASE24 Files Maintenance Manual***.

The TERM PROFILE field must match the SURCHARGE PROFILE configured on the appropriate ATD/TDF record(s).

The surcharge amount retrieved from the SURF is sent to the IAN in the Surcharge subfield in Format 07 of P-61 in the BIM request.

4. STARTUP AND CONTROL

This section provides information on the configuration of the NBGC Interchange Interface process. It also describes the processing involved for initialisation, network control commands, timers and the network management message flow between the NBGC Interchange Interface process and the IAN.

4.1 PROCESS CONFIGURATION

The following paragraphs contain an explanation on how the NBGC Interchange Interface process is configured. They provide information about the startup logic of the NBGC Interchange Interface process, the way in which the NBGC Interchange Interface process logs messages, and the impact of logging token data on disk usage.

4.1.1 Startup

In the Network Environment File (NEF), the STARTUP attribute for the PROCESS object type specifies the startup logic for an application process. Because of the length of time required for initialisation, the NBGC Interchange Interface process should be assigned a startup logic of AUTOMATIC in its process definition. AUTOMATIC means the NBGC Interchange Interface process is started automatically when the XPNET logical network is initiated. This ensures that the NBGC Interchange Interface process is ready when it receives its first transaction.

4.1.2 Log Messages

Log messages are routed to certain destinations, based on the routing code that appears in the log messages. The NBGC Interchange Interface process routes all of its log messages with a standard routing code of zero.

The NBGC Interchange Interface process generates log messages to assist network operators in performing their daily tasks. Log messages are unsolicited messages generated by BASE24 processes that provide information regarding the internal condition of the system. They are a network operator's primary link to what is actually occurring in the system. Log messages provide operators with information ranging from system status, such as the completion of a procedure, to more serious circumstances requiring operator intervention.

Records in the LMCF indicate the standard log messages that have been modified.

For a complete description of log messages output by the NBGC Interchange Interface, refer to Appendix C.

4.1.3 Tokens

Additional disk space may be required on the HP NonStop (formerly Tandem) for the Release 5.0 or above BASE24 database depending on the number of tokens being logged to the files.

The BASE24-REL parameter in the LCONF indicates the release level of the BASE24 Logical Network. Records in the TKN indicate the tokens that must be logged.

The Release 5.0 or above BASE24 user can indicate what tokens should be logged by the NBGC Interchange Interface to its Interface Log File (ILF). The user can also indicate to the Release 5.0 or above BASE24-atm Authorisation process what tokens should be logged to the Transaction Log File (TLF).

When deciding whether or not to log a token to any Release 5.0 or above log file, the user should consider the resulting length of the log record with the new token added. If logging a new token to a log file increases the length of a log file record so that fewer records can fit in the maximum HP NonStop (formerly Tandem) data block, more data blocks (i.e., more disk space) will be required to hold the same number of records. This will increase the user's disk space requirements.

For more details, refer to the ***BASE24 Tokens Manual***.

4.2 INITIALISATION

During initialisation processing, the NBGC Interchange Interface performs the steps listed below. If a problem occurs during initialisation, the NBGC Interchange Interface will log a message and abort.

- Initialise the process Trap Handler routines.
- Initialise communication with the appropriate XPNET process.
- Open the LCONF and process all LCONF Assigns and Parameters.
- Open the Log Message Configuration File (LMCF), read modified log messages and close the LMCF.
- Initialise the Event Log.
- Initialise the process' Timer Table.
- Opens the Interchange Configuration File (ICF) and read the appropriate record for configuration details (keyed by process name).
- Initialise the required Security Devices.
- Locate either the KEYF or KEY6, dependent on the contents of the DES-TRIPLE-SINGLE parameter, using the LCONF assign KEYF or KEY6.
- Open the Key File (KEYF or KEY6), and read the appropriate record for security details (keyed by process name).
- Build the Process Control Table (PCT) using information from the ICF and KEYF or KEY6.
- Calculate the initial Systems Trace Audit Number (STAN) and the initial Authorisation Identification Response (AIR) values.
- Determine the interchange's cutover time and the BASE24 posting date by comparing the current time with the LN-CUTOVER parameter value from the LCONF.
- Allocate and initialise the required Extended Memory segment(s).
- Open the Interchange Log File (ILF) for the previous and current day.
- Open the Store-and-Forward File (SAF).
- Initialise all default Bitmaps. The EMF is read and processed (overriding the default table entries). Note: the "1324" and "1334" message types are not supported by the EMF Requester.
- Open the Token File, and read the appropriate TKN records into memory.

- Update the KEYF/KEY6 and ICF records read earlier.
- Start a Performance timer, Cutover timer and Session timer.
- If required to Logon automatically (as specified by the ICF Auto-Signon-Start Flag), initiate Echo Testing and Probing on all stations.
- Start a MAC Reset timer.
- Generate the log message ‘PROCESS Initialised’.

4.3 NETWORK CONTROL COMMANDS

This section contains the description and entry syntax of all network control text commands that the NBGC Interchange Interface process supports.

The NBGC Interchange Interface process supports the following text commands:

LOGON	Initiates applications level communications with NBGC.
LOGOFF	Terminates applications level communications with NBGC.
STATUS	Displays the Interchange Interface status.
PERFORMANCE	Displays the Interchange Interface performance statistics.
WARMBOOT	Reinitialises the Interchange Interface.
TIMERS	Displays the Interchange Timer table.
MACING	Enables or Disables the use of Message Authentication Codes.
TRACE	Enables or disables the Trace facility.
DEBUG	Places the process into debug.
DUMP	Terminates the process and creates a dump file.

The above commands are issued to the NBGC Interchange Interface by means of the DELIVER PROCESS command. For example:

```
DELIVER PROCESS xxxxxxxx,'command'
```

where ‘xxxxxxxx’ represents the symbolic name of the NBGC Interchange Interface process.

For more details, refer to the ***NET24-XPNET Network Control Operations Guide*** and the ***NET24-XPNET Network Control Command Reference Manual.***

More details of the above commands are given on the following pages.

4.3.1 LOGON Command

The LOGON command causes the Interchange Interface to establish applications level communications with NBGC. The syntax of the command is:

```
DELIVER PROCESS xxxxxxxx,'LOGON'
```

The Interchange Interface commences echo testing on all stations. A message is logged stating that the Interface will logon when the link has been established. Once an echo test response has been received, the link is marked up and a logon message is formatted and sent to the IAN. If the KEY PROCESSING TYPE on the KEYF/KEY6 is set to 'S', then the logon message will not be sent to the IAN, and a log message will be generated.

The logon message exchanges the keys used to perform PIN block translation and message authentication. New session keys for PINs and MACs are exchanged at every logon. New transport keys, used to encrypt the session keys during transmission, are exchanged periodically, as specified by information held in the ICF.

Once the message is formatted and collapsed, a MAC is generated and appended to the message.

A log message indicating whether the logon was successful is generated.

If the Interchange Interface is already logged on, or is currently logging on or logging off, the LOGON command is ignored. A warning message is logged instead.

4.3.2 LOGOFF Command

The LOGOFF command causes the Interchange Interface to terminate applications level communication with NBGC. It is used to put the Interface into a Maintenance state in the event of database problems. The syntax of the command is:

```
DELIVER PROCESS xxxxxxxx,'LOGOFF'
```

A log message indicating that the Interchange Interface is logging off is generated. The logoff message includes a code indicating the reason for the logoff request; for an NCPCOM LOGOFF command, the reason code is ‘00’. When the logoff response is received from the IAN, all stations are logically marked down. This generates a log message for each station running to show that it has been marked down, and a log message stating that the link is marked down.

Note that no MAC is appended to a logoff request.

- If the Interchange Interface is already logged off, or is currently logging off, the LOGOFF command is ignored. A warning message is logged instead.
- If the Interface is currently logging on, the logon attempt is cancelled, and a warning message logged.

Once an Interchange Interface has been logged off by an NCPCOM command, it can only be logged on by means of the NCPCOM LOGON command. Request messages received from the IAN will be ignored. Acquired transactions destined for the IAN will be denied because the link is unavailable.

4.3.3 STATUS Command

The STATUS command causes the Interchange Interface to display details of the current Interface status in the BASE24 log. The syntax of the command is:

```
DELIVER PROCESS xxxxxxxx,'STATUS'
```

A short report is output in the log that includes the following details:

- The state of the Interchange, as detailed in the following table:

Status	Description
MAINT (TRUE or FALSE)	Indicates whether the Interface is prepared to attempt to establish application level communications. Note that FALSE is the operational state.
LINK (UP or DOWN)	Indicates whether any Interchange stations are started and operational.
CURRENT STATE (LOGGED ON, LOGGED OFF, LOGON PENDING, LOGOFF PENDING)	Indicates the current state of the Interface.
RETRY LOGON (TRUE or FALSE)	Indicates whether the Interface will logon automatically once the link has been re-established by echo testing.
TRANSPORT KEY EXCHANGE ENABLED or DISABLED	Indicates whether the exchange of new transported keys has been inhibited in the ICF.
MACING DISABLED	Indicates MACs are not to be added to any message.

- The Local Business Date.
- The number of inbound and outbound requests that are pending.
- The name and percentage full capacity of the SAF and ILF files used by the Interface.
- The status of all configured BASE24 stations used by the Interface.

4.3.4 PERFORMANCE Command

The PERFORMANCE command causes the Interchange Interface to display interface performance statistics in the BASE24 log. The syntax of the command is:

```
DELIVER PROCESS xxxxxxxx,'PERFORMANCE'
```

A short report is output to the log that includes the following details for the current and past performance periods:

- The number of outbound requests processed, their average response time and the number of time-outs.

- The number of inbound requests processed, their average response time and the number of time-outs.

4.3.5 WARMBOOT Command

The WARMBOOT command causes the NBGC Interchange Interface to reinitialise. The syntax of the command is:

```
DELIVER PROCESS xxxxxxxx,'WARMBOOT'
```

On completion of the re-initialisation processing, a log message is generated to indicate that it has warmbooted.

The WARMBOOT command is used when configuration parameters are changed and it is required that the Interchange should process according to the new configuration, for example, ICF parameters such as timer or error thresholds.

Note:

- Not all configuration changes will be acted upon as a result of a warmboot. For instance, it is not permissible to change station names using the WARMBOOT command.

- The values of the current state, the transport key exchange disabled, the retry logon, and the MACing disabled flag are not affected by the WARMBOOT command.

- The key exchange time from the ICF is updated in the PCT by a warmboot, but the new exchange time is not used until the current session timer expires.

4.3.6 TIMERS Command

The TIMERS command causes the NBGC Interchange Interface to display the type and duration of all outstanding timers on the BASE24 log. The syntax of the command is:

```
DELIVER PROCESS xxxxxxxx,'TIMERS'
```

The timer information displayed includes:

- Type
- Subtype
- Time remaining

4.3.7 MACING Command

The MACING command causes the NBGC Interchange Interface to reverse the state of the global flag used to indicate whether full message MACing is required. The syntax of the command is:

```
DELIVER PROCESS xxxxxxxx,'MACING'
```

When the process starts up, MACing is automatically enabled. By sending the MACING command, MACing is disabled. Re-sending the command will re-enable MACing. This feature is designed to help facilitate testing and should not be used in a live/production environment.

Whenever the MACING command is issued, two log messages are generated. The first reports that the command has been activated, and the second indicates the new status of the MACing flag.

4.3.8 TRACE Command

The TRACE command causes the NBGC Interchange Interface to enable or disable the TRACE facility. The syntax of the command is:

```
DELIVER PROCESS xxxxxxxx,'TRACE <parameter>'
```

Where the TRACE parameters are:

ON	Turn all trace bits on
OFF	Turn all trace bits off
4	Turn trace bit 4 (procedure level trace) on
5	Turn trace bit 5 (initialisation trace) on
10	Turn trace bit 10 (command trace) on
13	Turn trace bit 13 (route trace) on.

TRACE is a diagnostic tool and should not be operational in a live/production environment.

4.3.9 DEBUG Command

The DEBUG command causes the NBGC Interchange Interface to place itself into debug. The syntax of the command is:

```
DELIVER PROCESS xxxxxxxx,'DEBUG'
```

DEBUG is a diagnostic tool and should not be operational in a live/production environment.

4.3.10 DUMP Command

The DUMP command causes the NBGC Interchange Interface to terminate and create a dump file. The syntax of the command is:

```
DELIVER PROCESS xxxxxxxx,'DUMP'
```

DUMP is a diagnostic tool and should not be operational in a live/production environment.

4.4 NETWORK MANAGEMENT MESSAGE FLOWS

Network management messages (NMMs) are used to control the message processing within the network.

4.4.1 Application Level Messages

The NBGC Interchange Interface uses application-level network management messages to support the following functions:

- Echo Test
- Logon
- Logoff

Network management messages are also used to support dynamic key exchange.

The status of a BASE24 system (and its associated stations) connected to the IAN is defined by application states. Changes to the status of an individual station are notified to the NBGC Interchange Interface process by the command types documented in the table below.

Command Type	Action on Station
9510	Stopped
9511	Started
9512	Suspended
9513	Reinstated
9516	Not Being Polled
9517	Being Polled
9518	Connected
9519	Disconnected

Further information about network management message flows is given in section 4.4 and section 4.9 of the ***BIM Functional Specifications***.

4.4.2 Improved TCP/IP Session Management

For TCP/IP connections to the IAN, the NBGC Interchange Interface supports "socket probing", as described in the ***B.I.M. Functional Specifications, Addendum B, Improved TCP/IP Session Management.*** Socket probing is a "handshake" mechanism that enables the health of individual TCP sessions to be monitored. Conceptually, socket probing is positioned within the data communications layer.

To comply with the requirements of Addendum B, the XPNET configuration for each line on which a TCP/IP station is configured must have the FMM attribute set to 'ALL'.

The NBGC Interchange Interface monitors each TCP/IP station configured for traffic, so that any message received (except a Probe request) indicates a functioning station for the defined Probe time interval. If the NBGC Interchange Interface receives a Probe request message from the IAN, then the interface will reply with a Probe response message, if socket probing has been enabled.

If the NBGC Interchange Interface does not receive a message on a station during the defined Probe time interval, then it will send a Probe request to the station and await a Probe response to confirm that the socket is functioning. If the NBGC Interchange Interface still does not receive a message on a station during the next Probe time interval, then it will resend the Probe request to the station. In the event of no messages being received following the generation of a defined number of Probe requests, then the NBGC Interchange Interface first isolates the station, so that no application messages are sent to the station. If the situation persists, then the NBGC Interchange Interface disconnects the station. Note that the NBGC Interchange Interface does not assume that the socket is functioning correctly if only a Probe Request message is received on a station (because this indicates that the IAN is questioning the status of the link).

Socket probing is enabled if the SOCKET-PROBING parameter in the LCONF is set to a value of ON, and the probe time interval is configured using the PROBE-INTERVAL parameter in the LCONF. The number of Probe requests that can be sent without a response before a station is isolated is configured using the ISOLATE-MISSED-PROBES parameter in the LCONF. The number of Probe requests that can be sent without a response before a station is disconnected is configured using the MAX-MISSED-PROBES parameter in the LCONF.

Socket probing is disabled if the SOCKET-PROBING parameter in the LCONF is not configured or set to a value of OFF.

4.5 SECURITY PROCESSING

This section provides information on the security processing supported by the NBGC Interchange Interface process. It describes the dynamic key management protocol, and the BASE24 security utilities invoked to support the required processing.

4.5.1 Security Key Management

This section describes the online key management procedures used by the NBGC Interchange Interface. Four different keys are generated by the Interface via the configured HSM, and exchanged with the IAN during the logon process. The four keys are:

- PIN session key—encrypts PIN blocks for this logon session only.
- MAC session key—generates and verifies MACs for this logon session only.
- PIN transport key—encrypts the PIN session key in the logon message.
- MAC transport key—encrypts the MAC session key in the logon message.

New session keys for PINs and MACs are exchanged at every logon. New transport keys, used to encrypt the session keys during transmission, are exchanged periodically, as specified by the values of the fields under TRANSPORT KEY EXCHANGE on ICF screen 12.

New session and transport keys are stored initially in memory. The HSM returns two values, RVS and RVT, which are encrypted versions for insertion into the logon message. When an approved logon response is received from the IAN, the new transport keys are written to the KEYF or KEY6.

If the logon attempt fails, the old transport keys are retained and the new transport keys are not written to the KEYF or KEY6. The session keys are only retained in memory.

If new transport keys are exchanged, the count of the number of sessions the current transport keys have been used is reset to 1. Otherwise, the count is automatically incremented by 1. This count is stored on ICF screen 12 in the SESSIONS USED field.

The MAX SESSIONS field on ICF Screen 12 specifies the limit on sessions for which a given transport key can be used. When the session count equals or exceeds the specified limit, new transport keys are generated and used at the next logon session.

The exchange of transport keys can be enabled or disabled by setting the flag in the EXCHANGE KEYS field on ICF screen 12. This feature simplifies the offline exchange of the master keys from which the transport keys are generated. The master keys are stored in the KEYF or KEY6.

To ensure that keys are exchanged regularly even when a session is uninterrupted, a session timer is set. This timer is inserted during initialisation and expires at the time displayed in the DAILY KEY EXCHANGE TIME field on ICF screen 12. When the timer expires, a new timer is added and the Interface automatically logs off. When the logoff response is received, a logon request is generated automatically using new keys. There is a short delay of 1-2 minutes between logging off and logging back on. This is to ensure that the link is still intact before sending a new logon after logging off.

The key exchange time specified in the ICF can be updated via the AFT screen, and the Interface notified via a WARMBOOT command. Note that although the process is notified of the new exchange time, it will not use the new time to insert a session timer until the current session timer expires.

4.5.2 Security Utility Libraries

The BASE24 Release 6.0 HISWUTLS and SECUTILS modules support a number of functions implemented specifically for processing in the Dutch banking environment. The NBGC Interchange Interface invokes six of these functions, as shown in the table below.

HISWUTLS Procedure Name	SECUTILS Procedure Name
HISWSEC_ANSI_MAC_GENERATE	PIN_ANSI_MAC_GENERATE
HISWSEC_ANSI_MAC_VERIFY	PIN_ANSI_MAC_VERIFY
HISWSEC_GEN_RVS	PIN_GEN_RVS
HISWSEC_GEN_RVT	PIN_GEN_RVT
HISWSEC_RCV_RVS	PIN_RCV_RVS
HISWSEC_RCV_RVT	PIN_RCV_RVT

Each HISWUTLS procedure invokes the corresponding SECUTILS procedure. Each SECUTILS procedure either invokes a lower-level procedure that formats and sends the appropriate command to a Thales (formerly Racal) HSM or invokes a lower-level procedure that formats and sends the appropriate command message to TSS.

4.6 TIMERS

This section describes each timer used by the NBGC Interchange Interface. Each timer description includes the timer's function, and how and when the timer is set.

4.6.1 Outbound Requests Timer

An outbound request timer is set each time a 0100 or 0200 message is sent to the IAN. The number of seconds used for this timer is taken from the TIMER LIMITS - OUTBOUND field on ICF screen 8. The outbound request timer is used to indicate whether a transaction sent to the IAN has timed out.

If the timer expires before a response is received from the IAN for the 0100 or 0200 message, then the request is denied and the appropriate performance and timeout counters are incremented. If the expiration of this timer causes the maximum allowed timeouts to be reached, the connection to the IAN is marked down.

4.6.2 Inbound Requests Timer

An inbound request timer is set each time a 0200 message is sent to a BASE24 process for authorisation. The number of seconds used for this timer is taken from the TIMER LIMITS - INBOUND field on ICF screen 8. The inbound request timer is used to determine that an external transaction has timed out.

If the timer expires before a response is received from the authorising process, the request is denied. The appropriate performance and timeout counters are incremented.

4.6.3 Store-And-Forward Timer

A store-and-forward timer is set each time a store-and-forward message is sent to the IAN. The number of seconds used for this timer is taken from the TIMER LIMITS - STORE AND FORWARD field on ICF screen 8 for BASE24-atm transactions. The store and forward timer is used to determine that a Store-and-Forward File (SAF) record sent to the IAN has timed out.

If the timer expires before an acknowledgment is received from the IAN, the SAF timeout counter is incremented. If the SAF timeout counter is equal to the maximum number of retries set in the MAX SAF RETRY field on ICF screen 3, the record is written to the log and deleted from the SAF. If the MAX SAF RETRY field is set to 0, the SAF message will be sent indefinitely until it is successfully processed.

4.6.4 Network Management Timer

A network management timer is set each time a network management message is sent to a station. The number of seconds used for this timer is taken from the TIMER LIMITS - NETWORK MANAGEMENT field on ICF screen 3. The network management timer is used to monitor the communications between BASE24 and the IAN.

If the network management timer expires before a response is received from the IAN, an extended network management timer is set.

4.6.5 Extended Network Management Timer

An extended network management timer is set each time a network management timer expires without a response being received from the IAN. The number of seconds used for this timer is taken from the TIMER LIMITS - EXTENDED NETWORK field on ICF screen 3.

The network management message is repeated at the interval specified by the extended network management timer until an acknowledgment is received from the IAN or the communications line is marked up.

4.6.6 Wait For Traffic Timer

A wait for traffic timer is set each time a station is marked up and each time a previously set wait for traffic timer expires. The number of seconds used for this timer is taken from the TIMER LIMITS - WAIT FOR TRAFFIC field on ICF screen 3.

If the timer expires and message traffic has occurred on the line, a new wait for traffic timer is started.

If the timer expires and no message traffic has occurred on the line, a network management logon message is sent. The line is marked up or down, depending on the results of the logon.

4.6.7 Performance Timer

A performance timer is set at the start of each new performance interval. The number of minutes for this timer is taken from the TIMER LIMITS - PERFORMANCE PERIOD field on ICF screen 3. The performance timer triggers the end of one performance period and the beginning of the next.

When the performance timer expires, the NBGC Interchange Interface process:

- Moves the interval four performance statistics to interval five.
- Moves the interval three performance statistics to interval four.
- Moves the interval two performance statistics to interval three.
- Moves the interval one performance statistics to interval two.
- Initialises the interval one performance statistics to zero.
- Starts a new performance timer.

4.6.8 Cutover Timer

The cutover timer is used to ensure that the process cuts over to the next switch business day. It is set at initialisation and at interchange cutover. The value for this timer is taken from the SETTLEMENT HOUR and SETTLEMENT MINUTE fields on ICF Screen 2.

When the cutover timer expires, the SWITCH POSTING DATE on ICF screen 2 is updated to the next business date. A new ILF is created (if required) and opened, the old ILF is closed, and the LCONF parameter SW-ILF-FILE-RETENTION is checked to determine whether any ILFs should be deleted. A new cutover timer is started.

4.6.9 Session Timer

The session timer is used to ensure that session keys are exchanged with the IAN on a daily basis. It is set at initialisation and at session key exchange. The value for this timer is taken from the DAILY KEY EXCHANGE TIME field on ICF Screen 12.

When the session timer expires, session keys are exchanged with the IAN, and a new session timer is started.

4.6.10 MAC Reset Timer

The MAC reset timer is used to reset the bad MAC counter maintained by the NBGC Interchange Interface process. It is set at initialisation and when the bad MAC counter is reset. The value for this timer is taken from the NBGC-BAD-MAC-RESET-INTERVAL parameter in the LCONF.

When the MAC reset timer expires, the bad MAC counter is reset, and a new MAC reset timer is started.

4.6.11 Key Management Request Timer

A key management request timer is set each time an 0800 change key or change key request message is sent to a station. The number of seconds used for this timer is taken from the TIMER LIMITS NETWORK MANAGEMENT field on ICF screen 3.

If the network management timer expires before a response is received from the IAN, an extended key management request timer is set.

4.6.12 Extended Key Management Request Timer

An extended key management request timer is set each time a key management request timer expires without a response being received from the IAN. The number of seconds used for this timer is taken from the TIMER LIMITS EXTENDED NETWORK field on ICF screen 3.

The network management message is repeated at the interval specified by the extended key management timer until an acknowledgment is received from the IAN.

4.6.13 Probe Timer

A probe timer is set under the following scenarios:

- when an echo test is being sent on a station;
- when a station is started by the interface process;
- when the interface has determined the station on which a message (except a probe request) has been received;
- when a probe timer has expired and a message has been received on the station since the previous probe timer was set;
- when a probe request message is being sent to the station.

The value for this timer is taken from PROBE-INTERVAL parameter in the LCONF.

4.7 MESSAGE VALIDATION

This section provides information on the message validation supported by the NBGC Interchange Interface process.

4.7.1 Message Errors

The NBGC Interchange Interface validates all messages received from the IAN to ensure that they comply with the structure and content specified in the BIM protocol.

The NBGC Interchange Interface maintains a count of bad messages. This count is reset to zero when a valid message is received.

ICF Screen 12 contains the MAX BAD MESSAGES field. The NBGC Interchange Interface will send a LOGOFF to the IAN when the bad message counter exceeds this value. Manual intervention is required to warmboot and logon from NCPCOM.

4.7.2 MAC Errors

The NBGC Interchange Interface validates the MAC on all messages received from the IAN, if a MAC is present.

The NBGC Interchange Interface maintains a count of bad MACs. This count is reset to zero after every interval configured via the LCONF parameter NBGC-BAD-MAC-RESET-INTERVAL.

ICF Screen 12 contains the MAX BAD MACS field.

- If MAX BAD MACS is set to one, then the NBGC Interchange Interface will send a LOGOFF to the IAN when the first bad MAC occurs.
- If MAX BAD MACS is greater than one, then the NBGC Interchange Interface will perform a MAC key exchange with the IAN, when the count of bad MACS reaches one. This involves sending a LOGOFF, waiting for the next echo test, and then sending a LOGON. The LOGON message contains the new keys. If subsequent bad MACs occur (before the next bad MAC reset interval), and the bad MAC count reaches MAX BAD MACS, then a LOGOFF is sent to the IAN.

When a LOGOFF has been sent to the IAN, manual intervention is required to warmboot and logon from NCPCOM.

5. TRANSACTION MESSAGE HANDLING

The following functions related to transaction processing are described in this section:

- Transaction Message Flows
- Transaction Matching
- Transaction Security
- Transaction Types
- Interchange Log File (ILF) Processing
- Store-and-Forward (SAF) Processing
- Response Message Processing
- Reversal Message Processing

5.1 TRANSACTION MESSAGE FLOWS

Transaction messages are used to facilitate transaction processing within the network.

The NBGC Interchange Interface supports the following transaction message flows:

- Balance Inquiry dialogue
- Pre-authorisation dialogue
- Authorisation Request dialogue
- Authorisation Advice dialogue
- Acquirer Reversal dialogue.

Further information about transaction message flows is given in section 4.4 and section 4.9 of the ***BIM Functional Specifications***.

5.2 TRANSACTION MATCHING

In order to trace transactions, the NBGC Interchange Interface uses a combination of fields from the external message. The data elements from the external message that are used to trace transactions are listed below.

- Data element P-11 is the Systems Trace Audit Number
- Data element P-32 is the Acquiring Institution Identification Code
- Data element P-41 is the Card Acceptor Terminal Id
- Data element P-12 is the Local Transaction Time (BASE24-originated transactions only)
- Data element P-13 is the Local Transaction Time (BASE24-originated transactions only)

For reversal (0420) messages, the Systems Trace Audit Number and the Acquiring Institution Identification Code used for matching purposes are retrieved from field S-90 (sub-fields 2 and 4 respectively).

5.2.1 Definition of a Unique STAN

To ensure a unique Systems Trace Audit Number (STAN) is used when multiple NBGC Interchange Interface processes exist in a single BASE24 system, the LCONF parameter NBGC-STAN-PREFIX needs to be configured.

This parameter contains a unique value, which is used as the first digit in the generation of the Systems Audit Trace Number (STAN) used in external messages. All Issuer NBGC processes must have the value ‘0’ and each Acquirer NBGC process must have a unique value in the range 1 – 9. **A maximum of 9 NBGC Interchange Interface acquirer processes can be configured on the system.**

At initialisation, the STAN is read from the ICF (where it is stored during transaction processing), and the initial STAN value is set to PNNNNN; where ‘P’ is the value of this parameter and ‘NNNNN’ is the value read from the ICF plus 1000. The value of ‘NNNNN’ rolls over from 99999 to 1. If the STAN value is not in the range 1 - 99999, it will be set to 1. When the STAN value is calculated, this value on the ICF is updated.

To avoid duplicate STANs after a process failure, the STAN is written to the ICF after every 1000 transactions. On re-initialisation, the STAN is read from the ICF and incremented by 1000 to allow for the possibility of 999 transactions having occurred after the previous write of the STAN to the ICF, before the process failed.

A maximum of 99999 unique STANs per process per day is allowed.

5.3 TRANSACTION SECURITY

PIN blocks are encrypted using Triple DES in the interface with the IAN.

The processing of PIN blocks depends on whether the BASE24 system is supporting the "full" Triple DES solution (Transaction Security Services supported) or the "interim" Triple DES solution (Transaction Security Services not supported).

5.3.1 Triple DES "Full" Solution

This processing is applicable to BASE24 Release 6.0 systems only.

The double-length PIN Session Key is dynamically exchanged between the NBGC Interchange Interface and the interchange. This value is stored in memory.

A PIN-based transaction acquired by BASE24 and forwarded to the BASE24 NBGC Interchange Interface contains a PIN block encrypted under a key identified by a key locator in the BASE24 Standard Internal Message (STM). At the NBGC Interchange Interface, the PIN is translated from being encrypted under this key to being encrypted under the PIN Session Key. The transaction is then passed to the IAN.

For a PIN-based transaction request acquired by another Interpay member and received at the BASE24 NBGC Interchange Interface, the PIN block in the external message is encrypted under the PIN Session Key. At the NBGC Interchange Interface, the PIN and the key locator to the PIN Session Key (held in memory) are moved to the BASE24 Standard Internal Message (STM).

Further details of the TSS are given in the ***BASE24 Transaction Security Services Processing Guide***.

5.3.2 Triple DES "Interim" Solution

The double-length PIN Session Key is dynamically exchanged between the NBGC Interchange Interface and the interchange. This value is stored in memory. A single length intermediate PIN block encryption key (KPE) is stored on screen 1 of the KEY6.

A PIN-based transaction acquired by BASE24 and forwarded to the BASE24 NBGC Interchange Interface contains a PIN block encrypted under a single length key held in the BASE24 Standard Internal Message (STM). At the NBGC Interchange Interface, the PIN is translated from being encrypted under this key to being encrypted under the PIN Session Key. The transaction is then passed to the IAN.

For a PIN-based transaction request acquired by another Interpay member and received at the BASE24 NBGC Interchange Interface, the PIN block in the external message is encrypted under the PIN Session Key. At the NBGC Interchange Interface, the PIN block is translated from being encrypted under this key to being encrypted under the single length intermediate PIN block encryption key (KPE) from screen 1 of the KEY6. The PIN block and the single length Intermediate PIN block encryption key (KPE) are moved to the BASE24 Standard Internal Message (STM).

5.4 TRANSACTION TYPES

This section describes specific processing performed on transactions by the NBGC Interchange Interface.

5.4.1 Acquirer Processing

When a transaction message is sent to the IAN, the BASE24-atm transaction code is mapped to a BIM processing code, as shown below.

BASE24-atm Transaction Code		BIM Processing Code	
10	Cash Withdrawal	01	Cash Withdrawal
30	Balance Enquiry	31	Balance Enquiry

The “from” and “to” account types in the BIM processing code are both set to “00”.

Other BASE24-atm transaction types are mapped to a cash withdrawal transaction.

5.4.2 Issuer Processing

When a transaction message is received from the IAN, the BIM processing code is mapped to a BASE24-atm transaction, as shown below.

BIM Processing Code		BASE24-atm Transaction Code	
00	Goods or Service Purchase	10	Cash Withdrawal
01	Cash Transaction	10	Cash Withdrawal
07	Pre-Authorisation	30	Balance Enquiry
14	Purse Load	10	Cash Withdrawal
31	Balance Enquiry	30	Balance Enquiry

When a transaction message is received from the IAN, the BASE24-atm “from” account type is set to “01” (Checking) and the BASE24-atm “to” account type is set to “00” (No account needed).

Transaction messages containing any other processing code are rejected.

5.5 INTERCHANGE LOG FILE (ILF) PROCESSING

Transaction messages handled by the NBGC Interchange Interface are logged to the ILF.

5.5.1 Record Format

The transactions logged to the ILF by the NBGC Interchange Interface follow a standard structure that contains a fixed length portion (818 bytes), followed by any tokens configured for logging to the ILF.

Each ILF record has the following format:

Bytes 1–55	Primary Key
Bytes 56–98	ICF Record Information
Bytes 99–150	External Message Information
Bytes 151–818	STM Message Data
Bytes 819–4072	Token data (variable length depending on the token data)

The primary key for the NBGC ILF contains the following information:

- Data element P-32 is the Acquiring Institution Identification Code
- Data element P-41 is the Card Acceptor Terminal Id
- Data element P-11 is the Systems Trace Audit Number
- Data element P-12 is the Local Transaction Time
- Data element P-13 is the Local Transaction Time

The NBGC Interchange Interface provides users with the capability of identifying when the ILF is becoming full. The NBGC Interchange Interface maintains a counter of records written to the ILF. Each time the counter exceeds the value of the LCONF parameter ILF-NOTIFY-INTERVAL, the interface resets the counter and calculates how full the file has become. If the percentage of the file used exceeds the value of the LCONF parameter ILF-NOTIFY-PERCENTAGE, a warning message is logged.

5.6 STORE-AND-FORWARD (SAF) PROCESSING

The following is a description of the Store-and-Forward (SAF) processing performed by the NBGC Interchange Interface. Only reversal advices and transaction advices are placed on the SAF.

1. If a reversal or force post transaction (advice) is received from BASE24, an external message will be formatted and placed on the NBGC Interchange Interface SAF for processing.
2. When the NBGC Interchange Interface determines that there are transactions on the SAF (the SAF counter is greater than zero), the NBGC Interchange Interface will send an external reversal advice or force post (advice) transaction to the IAN. The ICF SAF timer is set in the NBGC Interchange Interface for an acknowledgment to the transaction.
3. The IAN will process the reversal advice or force post (advice) transaction and send an acknowledgment message to the BASE24 NBGC Interchange Interface. The NBGC Interchange Interface process will delete the transaction from the SAF.
4. If an acknowledgment is not received before the SAF acknowledgment timer expires at the BASE24 NBGC Interchange Interface, the Interface will attempt to send the transaction again until an acknowledgment is received or until the maximum number of SAF retries has been reached (specified in the ICF).

The NBGC Interchange Interface provides users with the capability of identifying when the SAF is becoming full. The NBGC Interchange Interface maintains a counter of records written to the SAF. Each time the counter exceeds the value of the LCONF parameter SAF-NOTIFY-INTERVAL, the interface resets the counter and calculates how full the file has become. If the percentage of the file used exceeds the value of the LCONF parameter SAF-NOTIFY-PERCENTAGE, a warning message is logged.

5.7 RESPONSE MESSAGE PROCESSING

This section describes the processing performed by the NBGC Interchange Interface on response messages.

5.7.1 Definition of a Unique AIR

To ensure a unique Authorisation Identification Response (AIR) is used when multiple NBGC Interchange Interface processes exist in a single BASE24 system, the LCONF parameter NBGC-AIR-FIELD-PREFIX needs to be configured.

This parameter stores the prefix value that each NBGC Interchange Interface process uses to compose the unique Authorisation Identification Response (AIR). The AIR uniquely identifies every authorisation response sent from the NBGC Interchange Interface process to the issuer, in a given period.

At initialisation, an AIR counter (a decimal value) is read from the ICF (where it is stored during transaction processing), and the initial AIR value is set to PAAAAA; where ‘P’ is the value of this parameter and ‘AAAAA’ is calculated from the AIR counter. Each character ‘A’ may take any alphanumeric value (0-9, A-Z), and so all calculations of ‘AAAAA’ must take into account that the value is represented in BASE 36. The AIR counter is increased by 1000, and converted to BASE 36 representation.

The AIR counter is incremented on every transaction, and used to calculate a new value of ‘AAAAA’. The AIR counter rolls over from 60466175 to 0. When the AIR value is calculated, the value of the AIR counter on the ICF is updated.

To avoid duplicate AIRs after a process failure, the AIR counter is written to the ICF after every 1000 transactions. On re-initialisation, the AIR counter is read from the ICF and incremented by 1000 to allow for the possibility of 999 transactions having occurred after the previous write of the AIR counter to the ICF, before the process failed.

A maximum of 60466176 (36 ** 5) unique AIRs per process per day is allowed.

5.7.2 Response Code Translation

When sending a response message to the IAN, the NBGC Interchange Interface maps the BASE24-atm internal response code to an ISO response code supported by the IAN. When receiving a response message from the IAN, the NBGC Interchange Interface maps the ISO response code used by the IAN to a BASE24-atm internal response code.

The response code mapping tables are maintained in a source file (RESPTBLS) separate to the main NBGC Interchange Interface source file (NBGCS). This makes it easier for users of the NBGC Interchange Interface to adjust the response code mappings, if required.

For a complete description of response code mappings performed by the NBGC Interchange Interface, refer to Appendix A.

5.8 REVERSAL MESSAGE PROCESSING

This section describes the processing performed by the NBGC Interchange Interface on reversal messages.

5.8.1 Acquirer Processing

When a reversal message is received from BASE24-atm, the NBGC Interchange Interface attempts to locate the original transaction message in the ILF.

- If the corresponding message can be located in the ILF, then an external reversal message is formatted and sent to the IAN.
- If the corresponding message cannot be located in the ILF, then the reversal message is added to the ILF and dropped.

5.8.2 Issuer Processing

When a reversal message is received from the IAN, the NBGC Interchange Interface attempts to locate the original transaction message in the ILF.

- If the corresponding message can be located in the ILF, then an internal reversal message is formatted and sent to the BASE24-atm Authorisation process.
- If the corresponding message cannot be located in the ILF, then the reversal message is added to the ILF and dropped.

5.8.2.1 Token Processing

The NBGC Reversal Token must be included in messages sent to the ATM Authorisation process. However, it is not permitted to add new tokens to a transaction message when a reversal is received (because the length of the ILF record could be increased).

Therefore, when a request message is received from the IAN, space is reserved in the ILF record for the NBGC Reversal Token, so that it can be added to the STM if the transaction is subsequently reversed.

5.8.2.2 Early Reversals

If a reversal cannot be matched with the original request in the ILF, then the message may be an “early reversal”. An “early reversal” is a 0420 message received from the IAN before the corresponding request/advice message.

When a transaction message is received from the IAN, the NBGC Interchange Interface will attempt to locate a corresponding “early reversal” on the current day’s ILF. If no corresponding “early reversal” can be found, then the NBGC Interchange Interface will use the NBGC-EARLY-REVERSAL-TIME parameter from the LCONF to determine whether it should also check the previous day’s ILF. This parameter indicates the time of day after which the NBGC Interchange Interface will stop looking in the previous day’s ILF for matching “early reversal” transactions.

6. FILE ACTION MESSAGE HANDLING

The following functions related to file action processing are described in this section:

- File Action Message Flows
- File Action Message Processing

6.1 FILE ACTION MESSAGE FLOWS

This section describes File Action message flows.

Further information about File Action message flows is given in section 4.7 and section 4.10 of the ***BIM Functional Specifications***.

6.1.1 File Action Message Received from the IAN

Figure 6-1 provides an overview of the message flow when a File Action advice message is received from the IAN.

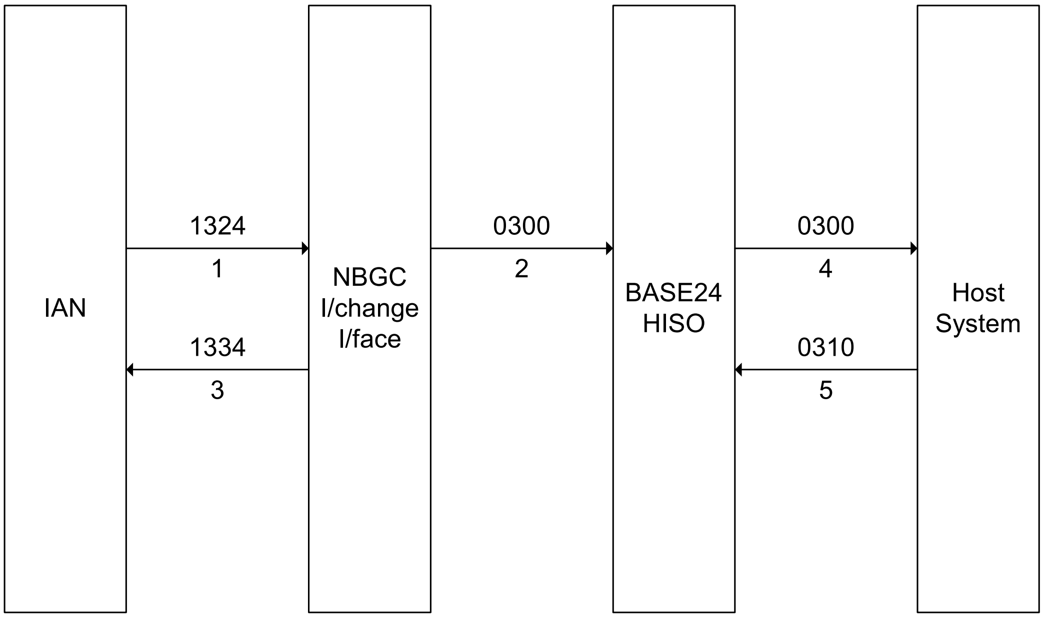

Figure 6-1 File Action Message Received from the IAN

- The NBGC Interchange Interface module receives the external 1324 message from the IAN.
- The NBGC Interchange Interface module formats the external advice message from the IAN into a 0300 SEM, and forwards the message to the Host Interface process.
- The NBGC Interchange Interface module formats an external acknowledgement message, and sends the 1334 message to the IAN.
- The Host Interface process receives the 0300 SEM, and forwards the message to the Host system.
- The Host system sends a 0310 SEM to the Host Interface module.

6.1.2 File Action Message Sent to the IAN

Figure 6-2 provides an overview of the message flow when a File Action advice message is sent to the IAN.

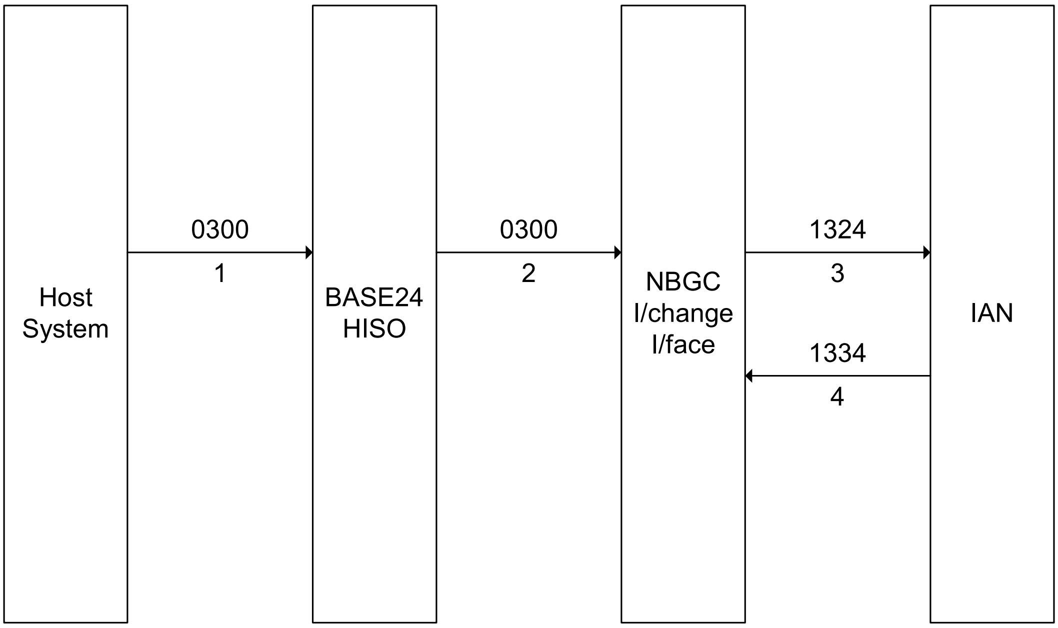

Figure 6-2 File Action Message Sent to the IAN

- The Host system sends a 0300 request message to the Host Interface module.
- The Host Interface module forwards the request message to the NBGC Interchange Interface process.
- The NBGC Interchange Interface module reformats the request message into an external message, and sends the 1324 advice message to the IAN.
- The NBGC Interchange Interface module receives a 1334 acknowledgement message from the IAN.

6.2 FILE ACTION MESSAGE PROCESSING

This section describes File Action message processing.

6.2.1 File Action Message Received from the IAN

The NBGC Interchange Interface receives a 1324 File Action advice from the IAN, and validates the message contents.

The NBGC Interchange Interface formats a SEM with a message type of 0300. The Data Processing Centre (DPC) number in the SEM is set to the value of the NBGC-BLOCKING-DPC parameter read from the LCONF.

The 0300 SEM is sent to the BASE24 process specified in the NBGC-BLOCKING-HOST parameter read from the LCONF.

The NBGC-BLOCKING-LOG-ILF parameter read from the LCONF is used to determine whether a record should be logged to the ILF.

The NBGC Interchange Interface formats a 1334 File Action advice acknowledgment, and sends the message to the IAN.

6.2.2 File Action Message Sent to the IAN

The NBGC Interchange Interface receives a 0300 SEM from the Host Interface process.

The NBGC Interchange Interface formats a 1324 File Action advice message. The Transaction Destination Institution Identification Code (field S-93) in the 1324 message is set to the value of the NBGC-BGC-BLOCKING-INST-ID parameter read from the LCONF. The Transaction Originator Institution Identification Code (field S-94) in the 1324 message is set to the value of the NBGC-B24-BLOCKING-INST-ID parameter read from the LCONF.

The 1324 message is added to the SAF file, and subsequently sent to the IAN.

The NBGC Interchange Interface receives a 1334 File Action advice acknowledgement from the IAN, and validates the message contents.

The NBGC-BLOCKING-LOG-ILF parameter read from the LCONF is used to determine whether a record should be logged to the ILF.

7. CUTOVER AND REPORTING

Within the BASE24 system, interchange cutover processing is a transaction management and reporting function that produces reports for each business day. These reports can be used in conjunction with BASE24-atm reports to reconcile the differences between the BASE24 processing day and the IAN processing day.

7.1 INTERCHANGE CUTOVER

The NBGC Interchange Interface cuts over to the next BASE24 business day when it receives a “9502” cutover command from the ATM Settlement process.

The NBGC Interchange Interface cuts over to the next switch business day when the cutover timer expires.

7.2 SETTLEMENT WITH INTERPAY

Settlement with Interpay is performed offline from BASE24 using extracts from transaction files (e.g., the ILF).

7.3 INTERCHANGE REPORTS

The BASE24 NBGC Interchange Interface cannot be configured to produce the standard BASE24 Interchange reports, but the standard BASE24 Interchange reports can be produced manually.

There are two reports of interchange transaction activity:

- SETL01 Clearings Statement Report
- STAT09 Transaction Detail Report

For details of the standard BASE24 Interchange reports, refer to the ***BASE24-atm Settlement and Reporting Manual***.

7.3.1 SETL01 Report

The Clearings Statement Report for the NBGC Interchange Interface reflects the debit, credit, and net position of the IAN to BASE24.

This report contains the totals of all transactions logged to the ILF.

7.3.2 STAT09 Report

The Transaction Detail Report provides a complete listing of all interchange activity. The report is based on the transactions logged to the current day's ILF.

This report contains one detail item for each transaction processed by the NBGC Interchange Interface during its processing day. The information provided for each transaction includes the cardholder number, transaction type, transaction time, posting date, and all transaction amounts. This report also includes a summary of the transaction detail information contained in the report.

To comply with the requirements of the PCI data security standards, some or all of the cardholder number in the report may be displayed masked with asterisks (*). The setting of the DATA MASK FLAG on Screen 2 of the ICF record determines whether the data in this field is masked. If masking is enabled for the interface process, then the values of the other three report data masking parameters on ICF Screen 2 determine which digits are masked.

7.3.3 Starting Reports

The BASE24 NBGC Interchange Interface cannot be configured to produce the standard BASE24 Interchange reports. However, the reports can be started manually by creating and running an obey file containing the necessary commands using Tandem Advanced Command Language (TACL).

If the report programs abend or fail, then they can be restarted in the same way.

8. EMV SUPPORT

The BASE24 NBGC EMV module is an additional licensable module, built on top of the Switch 6.0 version of the NBGC Interchange Interface.

The standard NBGC Interchange Interface source file includes a number of EMV-specific ‘hooks’ or user exits. These provide an interface between the standard NBGC Interchange Interface and the NBGC EMV module. The NBGC EMV module is bound together with the standard NBGC Interchange Interface code to provide support for BIM EMV transactions, in addition to non-EMV transactions.

8.1 BIM PROTOCOL

For EMV support, fields used in magnetic stripe transaction processing may contain different values, and fields not used in magnetic stripe transaction processing contain EMV-specific data. New message types have also been introduced to support EMV processing.

8.1.1 Fields Containing New Values for EMV Transactions

Fields used in magnetic stripe transaction processing contain different values to indicate that a transaction is being performed at a chip-capable terminal. However, chip data is included in authorisation messages only if the terminal has successfully read the chip.

For EMV transactions, the following new values in existing fields are supported:

- Values ‘05’ (PAN auto-entry via ICC), ‘80’ (PAN auto-entry via magnetic stripe – the full track data has been read and transmitted without alteration or truncation), and ‘95’ (PAN auto-entry via ICC; card data may be unreliable) are supported for digits 1 and 2 of the Point of Service Entry Mode (field P-22).

- Values ‘0’ (not a chip card), ‘1’ (“fallback” transaction - possible card problem), and ‘2’ (“fallback” transaction – possible terminal problem) are supported for the “Chip Condition Code” sub-field of the Additional Data, Private (field P-48.61).

- Values ‘0’ (non-chip transaction), and ‘1’ (chip transaction) are supported for the “Chip Transaction Indicator” sub-field of the Additional Data, Private (field P-48.62).

- Values ‘0’ (no problems experienced), and ‘1’ (card authentication may not be reliable) are supported for the “Authentication Reliability Indicator” sub-field of the Additional Data, Private (field P-48.63).

8.1.2 New Fields for EMV Transactions

Where a transaction is performed using a chip card and additional chip data is included in authorisation messages, extra fields are used in the BIM protocol.

Field P-23 contains the Application PAN Sequence Number.

Field P-55 contains the following sub-fields (identified by the appropriate EMV tags), required for the support of EMV transactions:

Tag	Name
9F5B	Issuer Script Results
4F	Application Identifier
71	Issuer Script Template 1
72	Issuer Script Template 2
82	Application Interchange Profile
84	Dedicated File Name
8A	Authorisation Response Code
91	Issuer Authentication Data
95	Terminal Verification Results
9A	Transaction Date
9C	Transaction Type
5F2A	Transaction Currency Code
9F02	Amount Authorised
9F03	Amount Other
9F08	Application Version Number
9F09	Terminal Application Version Number
9F10	Issuer Application Data
9F1A	Terminal Country Code
9F1E	Interface Device Serial Number
9F26	Application Cryptogram
9F27	Cryptogram Information Data
9F33	Terminal Capabilities
9F34	CVM Results
9F35	Terminal Type
9F36	Application Transaction Counter
9F37	Unpredictable Number
9F41	Transaction Sequence Counter
9F53	Transaction Category Code

In addition, a value of '5', '8', or '9' in position 11 of field P-63 indicates that the transaction originated from a terminal with an EMV-compatible ICC reader.

8.1.3 New Messages for EMV Transactions

To support the forwarding of informational advices generated by chip cards, extra message types are used in the BIM protocol.

The **Administrative Advice** (1624) is used when a chip card requires an informational advice to be sent to the card issuer, for example when a transaction is declined offline, or when issuer authentication fails, or when the results of issuer script processing are available. The **Administrative Advice Acknowledgement** (1634) is used to confirm receipt of the informational advice.

For administrative advices, new values in the following fields are supported:

- Function Code (field P-24);
- Action Code (field P-39);
- Data Record (field S-72).

For administrative advice acknowledgements, a new Action Code (field P-39) value is supported.

The NBGC Interchange Interface can process an administrative advice received from the IAN, and will respond with an administrative advice acknowledgement.

8.2 EMV TOKENS

EMV data processed by the BASE24 NBGC EMV module is passed within BASE24 in the form of data tokens appended to the STM. The following data tokens are used, dependent on the message type:

- **EMV Status Token (Token Id = ‘B4’).** This token contains data generated by the transaction acquirer and the acquiring interface. This data is used in authorisation processing for both EMV and non-EMV transactions, initiated at EMV-capable terminals.
- **EMV Request Token (Token Id = ‘B2’).** This token contains EMV data stored on the chip and generated by the terminal. This data is mandatory for EMV transactions, and is used in EMV transaction processing.
- **EMV Discretionary Token (Token Id = ‘B3’).** This token contains EMV data stored on the chip and generated by the terminal. This data is optional for EMV transactions, and is not used in EMV authorisation processing.
- **EMV Response Token (Token Id = ‘B5’).** This token contains EMV data generated by the card issuer. The EMV card uses this data to authenticate the card issuer.
- **EMV Script Token (Token Id = ‘B6’).** This token contains EMV data generated by the card issuer. The card issuer uses this data to re-configure the EMV card.
- **EMV Issuer Script Results Token (Token Id = ‘BJ’).** This token contains information about the processing of EMV Script data. The card issuer uses this information to determine the success or otherwise of an EMV Script sent to an EMV card.

8.3 BASE24 ACQUIRER PROCESSING

Where BASE24 is acting as the transaction acquirer (i.e. the card is issued by another institution), the NBGC EMV Module performs the processing described below.

8.3.1 Identifying EMV Transactions

There are, essentially, four levels of transaction support provided by the EMV product, based on the terminal capability, how the card details were read, and whether chip data elements are included in the transaction message.

On a request sent to the IAN, the four levels can be identified as described in the table below. Each level results in different processing requirements.

Terminal Capability	Card Details Entry	Chip Data Elements
EMV Status Token (indicating EMV-capable terminal) is not present (i.e. not EMV-capable).	Not applicable	Not applicable
EMV Status Token (indicating EMV-capable terminal) is present (i.e. EMV-capable).	POS Entry Mode in the token (indicating how the card details were read) is not set to '05x' (i.e. not chip-read).	Not applicable
EMV Status Token (indicating EMV-capable terminal) is present (i.e. EMV-capable).	POS Entry Mode in the token (indicating how the card details were read) is set to '05x' (i.e. chip-read).	EMV Request Token (indicating an EMV transaction) is not present (i.e. not an EMV transaction).
EMV Status Token (indicating EMV-capable terminal) is present (i.e. EMV-capable).	POS Entry Mode in the token (indicating how the card details were read) is set to '05x' (i.e. chip-read).	EMV Request Token (indicating an EMV transaction) is present (i.e. an EMV transaction).

8.3.2 Data Mapping

On a request sent to the IAN (an outbound message) and a response received from the IAN (an inbound message), EMV data is mapped as described below.

8.3.2.1 Outbound Messages

Where the transaction does not originate from an EMV-capable terminal (i.e. the first level of transaction support defined in the table in section 8.3.1), no additional EMV processing is performed.

Where the transaction does originate from an EMV-capable terminal, but the card details have not been read from the chip (i.e. the second level of transaction support defined in the table in section 8.3.1), data is mapped to the external message as shown in the table below.

External Message Field Name	Field Value
P-22 Point of Service Entry Mode	When the first 2 bytes of PT-SRV-ENTRY-MDE field in the EMV Status Token are “02” and the LAST-EMV-STAT field in the EMV Status Token contains “1” (i.e. an EMV card), then set the first 2 bytes of the field to “80” (i.e. a “fallback” transaction). Move the third byte of the PT-SRV-ENTRY-MDE into the third byte of the field.
P-48.61 Chip Condition Code	Move the value of the LAST-EMV-STAT field in the EMV Status Token into the field.
P-48.62 Chip Transaction Indicator	“0” (Non-chip transaction).

Where the transaction does originate from an EMV-capable terminal and the card details have been read from the chip, but EMV-specific data is not included in the authorisation message (i.e. the third level of transaction support defined in the table in section 8.3.1), the transaction will be declined. The BIM protocol does not permit this level of transaction support from an acquirer – if the card details are read from the chip, then the EMV-specific data must be included.

Where the transaction contains full EMV data (i.e. the fourth level of transaction support defined in the table in section 8.3.1), data is mapped to the external message as shown in the table below.

External Message Field Name	Field Value
P-22 Point of Service Entry Mode	When the AT50 Token is present and the COMPLETE-TRACK2-DATA field in the AT50 Token is “N”, then set the first 2 bytes of the field to “95”. When the AT50 Token is not present or the COMPLETE-TRACK2-DATA field in the AT50 Token is “Y”, then set the first 2 bytes of the field to “05”. Move the third byte of the PT-SRV-ENTRY-MDE into the third byte of the PSEM.ENTRY-MDE.
P-23 Card Sequence Number	If the APPL-PAN-SEQ-NUM field in the EMV Status Token does not contain spaces, then move the value (preceded by “0”) into the field.
P-48.62 Chip Transaction Indicator	“1” (Chip transaction).
P-48.63 Authentication Reliability Indicator	Move the value of the DATA-SUSPECT field in the EMV Status Token into the field.

In addition, data is mapped to field P-55 as shown in the table below.

P-55 Sub-Field Name	Sub-Field Value
Issuer Script Results	Moved from the EMV Issuer Script Results Token, if present.
Application Identifier	Not set.
Application Interchange Profile	Moved from the AIP field in the EMV Request Token.
Dedicated File Name	Moved from the DF-NAME field in the EMV Discretionary Token, if present.
Authorisation Response Code	Not set.
Terminal Verification Results	Moved from the TVR field in the EMV Request Token.
Transaction Date	Moved from the TRAN-DAT field in the EMV Request Token.

P-55 Sub-Field Name	Sub-Field Value
Transaction Type	Moved from the TRAN-TYPE field in the EMV Request Token.
Transaction Currency Code	Moved from the TRAN-CRNCY-CDE field in the EMV Request Token.
Amount Authorised	Moved from the AMT-AUTH field in the EMV Request Token.
Amount Other	Moved from the AMT-OTHER field in the EMV Request Token.
Application Version Number	Not set.
Terminal Application Version Number	Moved from the APPL-VER-NUM field in the EMV Discretionary Token, if present.
Issuer Application Data	Moved from the ISS-APPL-DATA field in the EMV Request Token, if present.
Terminal Country Code	Moved from the TERM-CNTRY-CDE field in the EMV Request Token.
Interface Device Serial Number	Moved from the TERM-SER-NUM field in the EMV Discretionary Token, if present.
Application Cryptogram	Moved from the ARQC field in the EMV Request Token.
Cryptogram Information Data	Moved from the CRYPTO-INFO-DATA field in the EMV Request Token.
Terminal Capabilities	Moved from the TERM-CAP field in the EMV Discretionary Token, if present.
CVM Results	Moved from the CVM-RSLTS field in the EMV Discretionary Token, if present.
Terminal Type	Moved from the EMV-TERM-TYPE field in the EMV Discretionary Token, if present.
Application Transaction Counter	Moved from the ATC field in the EMV Request Token.
Unpredictable Number	Moved from the UNPREDICT-NUM field in the EMV Request Token.
Transaction Sequence Counter	Not set.
Transaction Category Code	Not set.

8.3.2.2 Inbound Messages

Where the transaction response includes Issuer Authentication Data, the EMV Response Token is formatted as shown in the table below.

Response Token Field Name	Field Value
ISS-AUTH-DATA-LGTH	Set to the length component of tag ‘91’.
ISS-AUTH-DATA	Set to the value component of tag ‘91’.
SEND-CRD-BLK	“N” (Do not send Card Block script).
SEND-PUT-DATA	“N” (Do not send PUT DATA script).

Where the transaction response includes Issuer Script Template 1 and/or Issuer Script Template 2, the EMV Script Token is formatted as shown in the table below.

Script Token Field Name	Field Value
ISS-SCRIPT-DATA-LGTH	If Issuer Script Template 1 only is present, then set to the length component of tag ‘71’ plus 2 (to allow for the length of the tag and length components). If Issuer Script Template 2 only is present, then set to the length component of tag ‘72’ plus 2 (to allow for the length of the tag and length components). If both Issuer Script Template 1 and Issuer Script Template 2 are present, then set to the length component of tag ‘71’ plus 2 (to allow for the length of the tag and length components) plus the length component of tag ‘72’ plus 2 (to allow for the length of the tag and length components). If this value exceeds 128, then set to 128 instead.
ISS-SCRIPT-DATA	If Issuer Script Template 1 only is present, then set to the data in tag ‘71’ (including the tag and length components). If Issuer Script Template 2 only is present, then set to the data in tag ‘72’ (including the tag and length components). If both Issuer Script Template 1 and Issuer Script Template 2 are present, then set to the data in tag ‘71’ (including the tag and length components) concatenated with the data in tag ‘72’ (including the tag and length components). If the total data length exceeds 128 bytes, then truncate at 128 bytes.

8.4 BASE24 ISSUER PROCESSING

Where BASE24 is acting as the card issuer (i.e. the transaction is acquired by another institution), the NBGC EMV Module performs the processing described below.

8.4.1 Identifying EMV Transactions

There are, essentially, four levels of transaction support provided by the EMV product, based on the terminal capability, how the card details were read, and whether chip data elements are included in the transaction message.

On a request received from the IAN, the four levels can be identified as described in the table below. Each level results in different processing requirements.

Terminal Capability	Card Details Entry	Chip Data Elements
Field P-63 (POS Data) position 11 is not 5, 8 or '9' (i.e. not EMV capable).	Not applicable	Not applicable
Field P-63 (POS Data) position 11 is 5, 8 or '9' (i.e. EMV capable).	Field P-22 (POS Entry Mode) is not '05x' or '95x' (i.e. not read from chip)	Not applicable
Field P-63 (POS Data) position 11 is 5, 8 or '9' (i.e. EMV capable).	Field P-22 (POS Entry Mode) is '05x' or '95x' (i.e. read from chip)	Field P-55 (ICC System-Related Data) is not present (i.e. no EMV data)
Field P-63 (POS Data) position 11 is 5, 8 or '9' (i.e. EMV capable).	Field P-22 (POS Entry Mode) is '05x' or '95x' (i.e. read from chip)	Field P-55 (ICC System-Related Data) is present (i.e. EMV data)

8.4.2 Data Mapping

On a request received from the IAN (an inbound message) and a response sent to the IAN (an outbound message), EMV data is mapped as described below.

8.4.2.1 Inbound Messages

Where the transaction does not originate from an EMV-capable terminal (i.e. the first level of transaction support defined in the table in section 8.4.1), no additional EMV processing is performed.

Where the transaction does originate from an EMV-capable terminal, but the card details have not been read from the chip (i.e. the second level of transaction support defined in the table in section 8.4.1), the EMV Status Token is formatted as shown in the table below.

Status Token Field Name	Field Value
PT-SRV-ENTRY-MDE	If the first 2 characters of P-22 (POS Entry Mode) are not “80” or “90”, then set the first 2 bytes of the field to this value. If the first 2 characters of P-22 (POS Entry Mode) are “80” or “90”, then set the first 2 bytes of the field to “02”. Move the third byte of P-22 (POS Entry Mode) into the third byte of the field.
TERM-ENTRY-CAP	“5” (ICC read capability)
LAST-EMV-STAT	If P-48.61 (Chip Condition Code) is “2” or “1”, then set the field to “1” (chip card); otherwise set it to “0” (not a chip card).
DATA-SUSPECT	Not set.
APPL-PAN-SEQ-NUM	Not set.
DEV-INFO	Not set.
RSN-ON-LINE-CDE	Not set.
ARQC-VERIFY	“0” (ARQC not verified)
USER-FLD1	Not set.

Where the transaction does originate from an EMV-capable terminal and the card details have been read from the chip, but EMV-specific data is not included in the authorisation message (i.e. the third level of transaction support defined in the table in section 8.4.1), the EMV Status Token is formatted as shown in the table below.

Status Token Field Name	Field Value
PT-SRV-ENTRY-MDE	Set the first 2 bytes of the field to “05”. Move the third byte of P-22 (POS Entry Mode) into the third byte of the field.
TERM-ENTRY-CAP	“5” (ICC read capability)
LAST-EMV-STAT	“1” (Chip card).
DATA-SUSPECT	Move the value of P-48.63 (Authentication Reliability Indicator) into the field.
APPL-PAN-SEQ-NUM	Set the field to the last 2 bytes of P-23 (Card Sequence Number) if present; otherwise set the field to spaces.
DEV-INFO	Not set.
RSN-ON-LINE-CDE	Not set.
ARQC-VERIFY	“0” (ARQC not verified)
USER-FLD1	Not set.

Where the transaction contains full EMV data (i.e. the fourth level of transaction support defined in the table in section 8.4.1), the EMV Status Token is formatted as specified in the table above.

In addition, the EMV Request Token is formatted as shown in the table below.

Request Token Field Name	Field Value
BIT-MAP	Set according to the content of the subsequent fields. The ISO 8583 convention is followed, in that the highest order bit represents the first field in the token (following the BIT-MAP).
CRYPTO-INFO-DATA	Moved from the value component of the Cryptogram Information Data sub-field in P-55.
TVR	Moved from the value component of the Terminal Verification Results sub-field in P-55.
ARQC	Moved from the value component of the Application Cryptogram sub-field in P-55.
AMT-AUTH	Moved from the value component of the Amount Authorised sub-field in P-55.
AMT-OTHER	Moved from the value component of the Amount Other sub-field in P-55.
AIP	Moved from the value component of the Application Interchange Profile sub-field in P-55.
ATC	Moved from the value component of the Application Transaction Counter sub-field in P-55.
TERM-CNTRY-CDE	Moved from the value component of the Terminal Country Code sub-field in P-55.
TRAN-CRNCY-CDE	Moved from the value component of the Transaction Currency Code sub-field in P-55.
TRAN-DAT	Moved from the value component of the Transaction Date sub-field in P-55.
TRAN-TYPE	Moved from the value component of the Transaction Type sub-field in P-55.
UNPREDICT-NUM	Moved from the value component of the Unpredictable Number sub-field in P-55.
ISS-APPL-DATA-LGTH	Moved from the length component of the Issuer Application Data sub-field in P-55, if present.
ISS-APPL-DATA	Moved from the value component of the Issuer Application Data sub-field in P-55, if present.

In addition, the EMV Discretionary Token is formatted as shown in the table below, if any of the fields used to format the token are present in the incoming message.

Discretionary Token Field Name	Field Value
BIT-MAP	Set according to the content of the subsequent fields. The ISO 8583 convention is followed, in that the highest order bit represents the first field in the token (following the BIT-MAP).
TERM-SERL-NUM	Moved from the value component of the Interface Device Serial Number sub-field in P-55, if present.
EMV-TERM-CAP	Moved from the value component of the Terminal Capabilities sub-field in P-55, if present.
EMV-TERM-TYPE	Moved from the value component of the Terminal Type sub-field in P-55, if present.
APPL-VER-NUM	Moved from the value component of the Terminal Application Version Number sub-field in P-55, if present.
CVM-RSLTS	Moved from the value component of the Cardholder Verification Method Results sub-field in P-55, if present.
DF-NAME-LGTH	Moved from the length component of the Dedicated File Name sub-field in P-55, if present. Otherwise, moved from the length component of the Application Identifier sub-field in P-55, if present.
DF-NAME	Moved from the value component of the Dedicated File Name sub-field in P-55, if present. Otherwise, moved from the value component of the Application Identifier sub-field in P-55, if present.

Issuer Script Results are sent in reversal advices and administrative advices to inform the issuer of the cause of the script processing failure on the IC card. For reversal advices, this data is moved to the EMV Issuer Script Results token.

8.4.2.2 Outbound Messages

Where the transaction response includes the EMV Response Token, data is mapped to field P-55 as shown in the table below.

P-55 Sub-Field Name	Field Value
Issuer Authentication Data	Moved from the ISS-AUTH-DATA field in the EMV Response Token.

Where the transaction response includes the EMV Script Token, data is mapped to field P-55 as shown in the table below.

P-55 Sub-Field Name	Field Value
Issuer Script Template 1	Moved from the ISS-SCRIPT-DATA field in the EMV Script Token, if the first byte of the token data contains '71'. Otherwise, moved from the ISS-SCRIPT-DATA field in the EMV Script Token if the nth byte of the token data contains '71', where n = value of the second byte of the token data + 3 (i.e. an Issuer Script Template 1 has been concatenated with an Issuer Script Template 2).
Issuer Script Template 2	Moved from the ISS-SCRIPT-DATA field in the EMV Script Token, if the first byte of the token data contains '72'. Otherwise, moved from the ISS-SCRIPT-DATA field in the EMV Script Token if the nth byte of the token data contains '72', where n = value of the second byte of the token data + 3 (i.e. an Issuer Script Template 2 has been concatenated with an Issuer Script Template 1).

A RESPONSE CODE TRANSLATION

A.1 BASE24-atm TO NBGC RESPONSE CODE

BASE24-atm RESPONSE CODE		NBGC RESPONSE CODE	
00	Approved	00	Approved
01	Approved – no balances	00	Approved
50	Unauthorised usage	12	Card blocked, return card
51	Expired card	54	Expired card
52	Invalid card	14	Card not known to issuer
53	Incorrect PIN	55	Wrong PIN
54	BASE24 database problem	06	Declined, reason not specified
55	Ineligible transaction	06	Declined, reason not specified
56	Ineligible account	06	Declined, reason not specified
57	Transaction not supported	06	Declined, reason not specified
58	NSF – no available balance	51	Insufficient funds
59	NSF – w/ available balance	51	Insufficient funds
60	No further withdrawals	05	(Guest) usage not possible today
61	Enter lesser amount	61	Exceeds withdrawal limit amount
62	PIN tries exceeded	75	Allowable number of PIN tries exceeded, return card
63	Daily withdrawal limit reached	61	Exceeds withdrawal limit amount
64	Inv amt – CCD request	06	Declined, reason not specified
65	No statement print info	06	Declined, reason not specified
66	No statement print info	06	Declined, reason not specified
67	Invalid cash back amount	06	Declined, reason not specified
68	External decline	06	Declined, reason not specified
69	No sharing with issuer	57	Restricted card, transaction not permitted to cardholder
70	System error	06	Declined, reason not specified
71	Contact bank	06	Declined, reason not specified

BASE24-atm RESPONSE CODE		NBGC RESPONSE CODE	
72	Destination not available	06	Declined, reason not specified
73	Routing lookup problem	92	Unable to route transaction
74	Message edit error	30	Format error
80	Signature check failed	76	Wrong PIN, one attempt left
81	HSM parameter error	06	Declined, reason not specified
82	HSM failure	06	Declined, reason not specified
83	KEYI not found	06	Declined, reason not specified
84	ATC check failure	06	Declined, reason not specified
85	CVR check failure	06	Declined, reason not specified
86	TVR check failure	06	Declined, reason not specified
87	ARQC verification failure	06	Declined, reason not specified
88	Fallback transaction	06	Declined, reason not specified

If the first byte of the BASE24-atm response code is 1 (Capture), then the external response code is set to 04 (Card blocked, pick up card).

If the BASE24-atm response code is 53 (Incorrect PIN) and the ERR-FLG in the AT50 Token is ‘O’ (One Try Remaining), then the external response code is set to 76 (Wrong PIN, one attempt left).

If the BASE24-atm response code is 74 (Message edit error) and the CRD-VRFY-FLG in the AT50 Token is ‘D’ (Card Verification failed - Decline), then the external response code is set to 78 (Suspected counterfeit card).

Any BASE24-atm response codes not listed are translated to 06 (Declined, reason not specified).

A.2 NBGC TO BASE24-atm RESPONSE CODE

NBGC RESPONSE CODE		BASE24-atm RESPONSE CODE	
00	Approved	000	Approved
01	Refer to card issuer	071	Contact bank
04	Card blocked, pick up card	150	Unauthorised usage
05	(Guest) usage not possible today	060	No further withdrawals
06	Declined, reason not specified	054	BASE24 database problem
12	Card blocked, return card	050	Unauthorised usage
14	Card not known to issuer	052	Invalid card
30	Format error	074	Message edit error
34	Card has been blocked, pick up card	150	Unauthorised usage
40	Requested function not supported	072	Destination not available
51	Insufficient funds	058	NSF – no available balance
52	Insufficient funds	068	External decline
54	Expired card	051	Expired card
55	Wrong PIN	053	Incorrect PIN
57	Restricted card, transaction not permitted to cardholder	069	No sharing with issuer
58	Restricted card, transaction not permitted to terminal	057	Transaction not supported
61	Exceeds withdrawal limit amount	063	Daily withdrawal limit reached
65	Transaction denied, pay otherwise	054	BASE24 database problem
75	Allowable number of PIN tries exceeded, return card	062	PIN tries exceeded
76	Wrong PIN, one attempt left	053	Incorrect PIN
78	Suspected counterfeit card	074	Message edit error
91	Issuer or stand-in not available	072	Destination not available
92	Unable to route transaction	073	Routing lookup problem

Any codes not listed are translated to the BASE24-atm response code 068 (External Decline).

B FILE FORMATS

B.1 INTERCHANGE CONFIGURATION FILE (ICF)

This section gives a description of the fields included in the NBGC Interchange Interface specific segment of the ICF.

The NBGC-Specific ICF DDL contains one field for each piece of information recorded on ICF screen 12.

This structure is used to redefine the 400 bytes of ICF Segment 31 (Switch Specific Data).

Total definition size is 92 bytes.

```
DEFINITION ICFNBGC.

02   CO-NET-STATION                 OCCURS 4 TIMES.
   04   NAME                        PIC X(16).
02   NUM-STATIONS                   TYPE BINARY 16.
02   KEY-EXCHANGE-TIME              PIC X(4).
02   TRAN-KEY-EXCH.
   04   TRANSPORT-KEY-MAX           PIC 9(3).
   04   TRANSPORT-KEY-USAGE         PIC 9(3).
   04   TRANSPORT-KEY-EXCHANGE      PIC X.
02   RETRIES.
   04   MAX-BAD-MSG                 PIC 9(4).
   04   MAX-VERIFY-MACS             PIC 9(4).
02   USER-FLD1                      PIC X.
02   NBGC-STAN                      TYPE BINARY 32.
02   AIR-FIELD-CNTR                 TYPE BINARY 32.

END.
```

B.2 BASE24 GENERIC SWITCH TOKEN

The fields in the NBGC version of the Switch Token are described below.

LEVEL	FIELD NAME AND DESCRIPTION	PICTURE
02	LGTH	TYPE BINARY 16
02	FIID	PIC X(4)
02	BUF	PIC X(96)
02	NBGC	REDEFINES BUF
04	VER-ID The version number of the token	PIC X(2)
04	MSG-TYP The Message Type	PIC 9(4)
04	PROC-CDE P-3 - Processing Code	PIC 9(6)
04	STAN P-11 - System Trace Audit Number	PIC 9(6)
04	EXP-DAT P-14 – Expiration Date	PIC 9(4)
04	ACQ-CNTRY-CDE P-19 – Acquiring Institution Country Code	PIC 9(3)
04	POS-ENTRY-MDE P-22 - Point of Service Entry Mode	PIC 9(3)
04	POS-COND-CDE P-25 - Point of Service Condition Code	PIC 9(2)
04	FWD-INST-ID-CDE P-33 - Forwarding Institution ID Number	PIC 9(11)
04	RET-REF-NUM P-37 - Retrieval Reference Number	PIC X(12)
04	RESP-CDE P-39 - Response Code	PIC X(2)
04	CRD-ACPT-ID-CDE P-42 – Card Acceptor Identification Code	PIC X(15)

LEVEL	FIELD NAME AND DESCRIPTION	PICTURE
04	POS-DATA P-63 – POS Data	PIC X(26)
06	COND-CDE	PIC 9(11)
06	AUTH-LIFE-CYCLE	PIC 9(2)
06	CNTRY-CDE	PIC 9(3)
06	POSTAL-CDE	PIC X(10)

The FIID is hard-coded as ‘NBGC’.

B.3 NBGC TOKENS

The NBGC Interchange Interface supports the Dutch-specific tokens listed below.

NBGC-TRAN-TKN (X0)

The NBGC-TRAN-TKN (X0) product token (previously the NBGC-TLF-LOG-TKN) is used to specify critical transaction data that must be logged to the TLF.

Total definition size is 12 bytes.

```
DEFINITION NBGC-TRAN-TKN.

02  TRACK-FLAG                          PIC X.
02  SYSTEM-TRACE                        PIC X(6).
02  EXP-DATE                            PIC X(4).
02  TRAN-TYPE-FLAG                      PIC X.

END.
```

The fields are described below.

FIELD NAME	DESCRIPTION
TRACK-FLAG	A flag used to indicate which track is used for authorisation. Valid values are: "2" = Track-2 used for authorisation; "3" = Track-3 used for authorisation. For acquired transactions, this field is set by the Device Handler. For transactions received from the IAN, this field is set by the NBGC Interchange Interface.
SYSTEM-TRACE	The Systems Trace Audit Number (STAN). For acquired transactions, this field is set to ASCII zeros. For transactions received from the IAN, this field is set to the contents of BIM field P-11.
EXP-DATE	The expiration date of the card (YYMM). For acquired transactions, this field is set to ASCII zeros. For transactions received from the IAN, this field is set to the contents of BIM field P-14 (if present).

FIELD NAME	DESCRIPTION
TRAN-TYPE-FLAG	A flag used to indicate the type of transaction. Valid values are: "0" = on-us ATM transaction; "1" = foreign (not-on-us) ATM transaction; "2" = POS transaction; "3" = pre-auth transaction; "4" = chip card upload. For acquired transactions, this field is set to “0”. For transactions received from the IAN, this field is set by the NBGC Interchange Interface.

NBGC-RVSL-TKN (X1)

The NBGC-RVSL-TKN (X1) product token is used to indicate whether a reversal has been received for a ‘0220’ Completion message.

Total definition size is 2 bytes.

```
DEFINITION NBGC-RVSL-TKN.

02  RVSL-OF-0220                    PIC X.
02  USER-FLD                        PIC X.

END.
```

The NBGC Interchange Interface sets the RVSL-OF-0220 flag to "1" if a reversal for an 0220 message is received from the IAN. Otherwise, the NBGC Interchange Interface sets the RVSL-OF-0220 flag to "0".

The Dutch version of the BASE24 ATM Authorisation process uses this flag for two purposes:

- The BASE24-atm Authorisation process will not reformat the member number in the STM for reversals to 0220 messages. The NBGC Interchange Interface will have already formatted the member number. There may not be sufficient track data for the Authorisation process to format the member number.
- The BASE24-atm Authorisation process will not update PBF CASH OUT TODAY for a reversal to an 0220 message.

Note that if the token is not present on a reversal, then the BASE24 ATM Authorisation process assumes that the message is NOT a reversal of an 0220.

NBGC-PAN-TKN (X2)

The NBGC-PAN-TKN (X2) product token is used to contain the full PAN, which could potentially be a 22-digit extended PAN. This token is formatted by the appropriate Authorisation process for logging to the TLF.

Total definition size is 22 bytes.

```
DEFINITION NBGC-PAN-TKN.

02  PAN                                PIC X(22).

END.
```

C LOG MESSAGES

Log messages are written to the Tandem Event Management System (EMS) logging facility.

Log messages are generated for an error condition or informational purpose. Erroneous messages require corrective action before the Interchange Interface can function properly.

C.1 FORMAT OF MESSAGES

A typical format for BASE24 log messages is:

YY/MM/DD HH:MM:SS (X) <PROCESS ID> <OBJECT> MMMM RRRR <TEXT>

where: '`YY/MM/DD`' specifies the date the message was sent.

'`HH:MM:SS`' specifies the time the message was sent.

'`(X)`' specifies the severity code where:

(0) or (I)	Status (Informational)
(1) or (W)	Warning
(2) or (E)	Error
(3) or (C)	Critical

'`<PROCESS ID>`' specifies the symbolic process name.

'`<OBJECT>`' specifies a 16-byte string defined and passed by the process.

'`MMMM`' specifies the message number.

'`RRRR`' specifies the routing code.

'`<TEXT>`' specifies the text of the message.

C.2 LIST OF MESSAGES

The NBGCLOGM file on the SW60NBGC subvolume documents the log messages output by the NBGC Interchange Interface.

For each log message, additional information is given:

- the cause or reason the message is displayed
- the remedy or solution for the message, if relevant

Also, the following support/service personnel may need to be contacted:

- Operations Personnel

  In-house operations staff who are responsible for initiating fault action for a particular problem.

- HELP24

  ACI Worldwide personnel who are responsible for solving customer queries and problems.

D GLOSSARY

AFT	Application Files Terminal
AIR	Authorisation Identification Response
ATM	Automated Teller Machine
BIM	Bank Interchange Message
DDL	Data Definition Language
DES	Data Encryption Standard
EFT/POS	Electronic Funds Transfer/Point of Sale
EMS	Event Management System
EMV	Europay/MasterCard/Visa
HSM	Hardware Security Module
IAN	Interbancair Autorisatie Netwerk
ICC	Integrated Chip Card
ISO	International Organisation for Standardisation
MAC	Message Authentication Code
MOTO	Mail Order/Telephone Order
NBGC	Nederland BankGiroCentrale
NMM	Network Management Message
PAN	Primary Account Number
PCI	Payment Card Industry
PIN	Personal Identification Number
STAN	Systems Trace Audit Number
STM	Standard Internal Message
TSS	Transaction Security Services

ACI Worldwide (EMEA) Limited